

Biology for Engineers

Course Code: BIO 3105/3107

Introduction

Lecture-01

01/03/2026

Reference book: Biology for Engineers - Arthur T. Johnson [2nd edition]

Biology for Engineers – G. K. Suraishkumar

SM Rafiqul Islam, Ph.D.

Associate Professor and Head,

Biotechnology & Genetic Engineering (BGE) Department

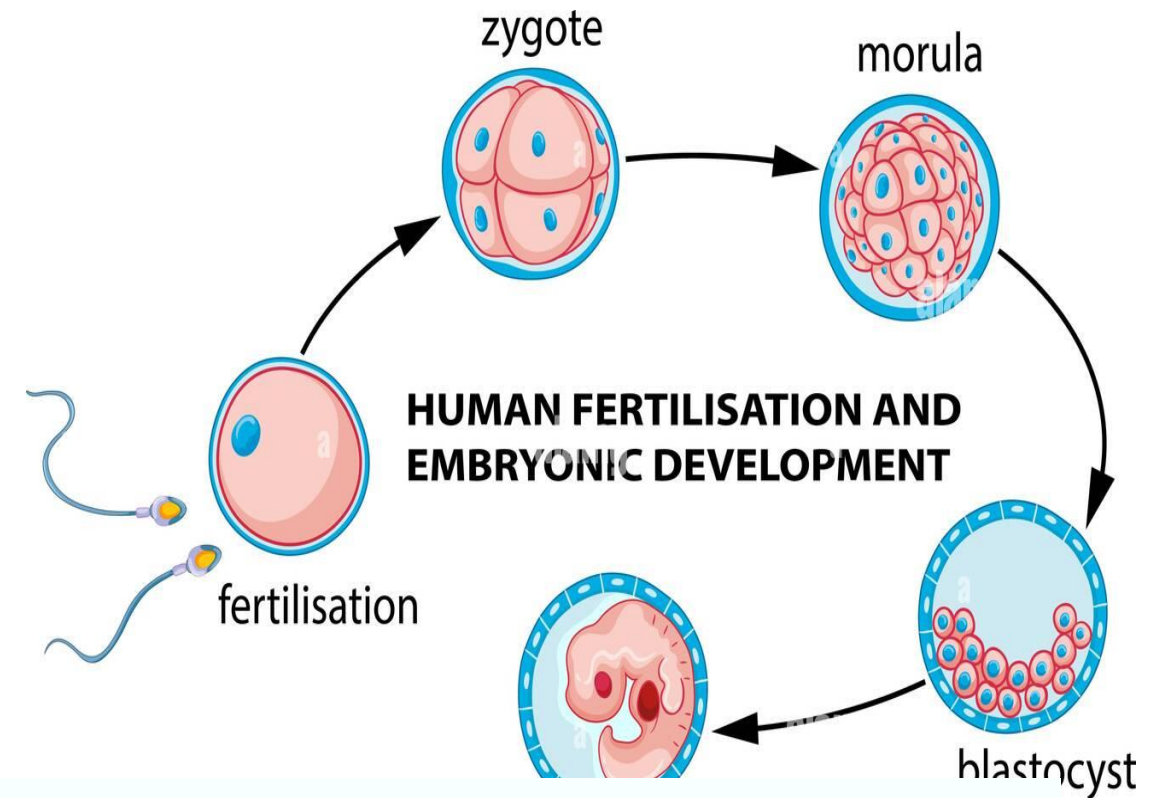
United International University

Introduction: Why Biology in Engineering

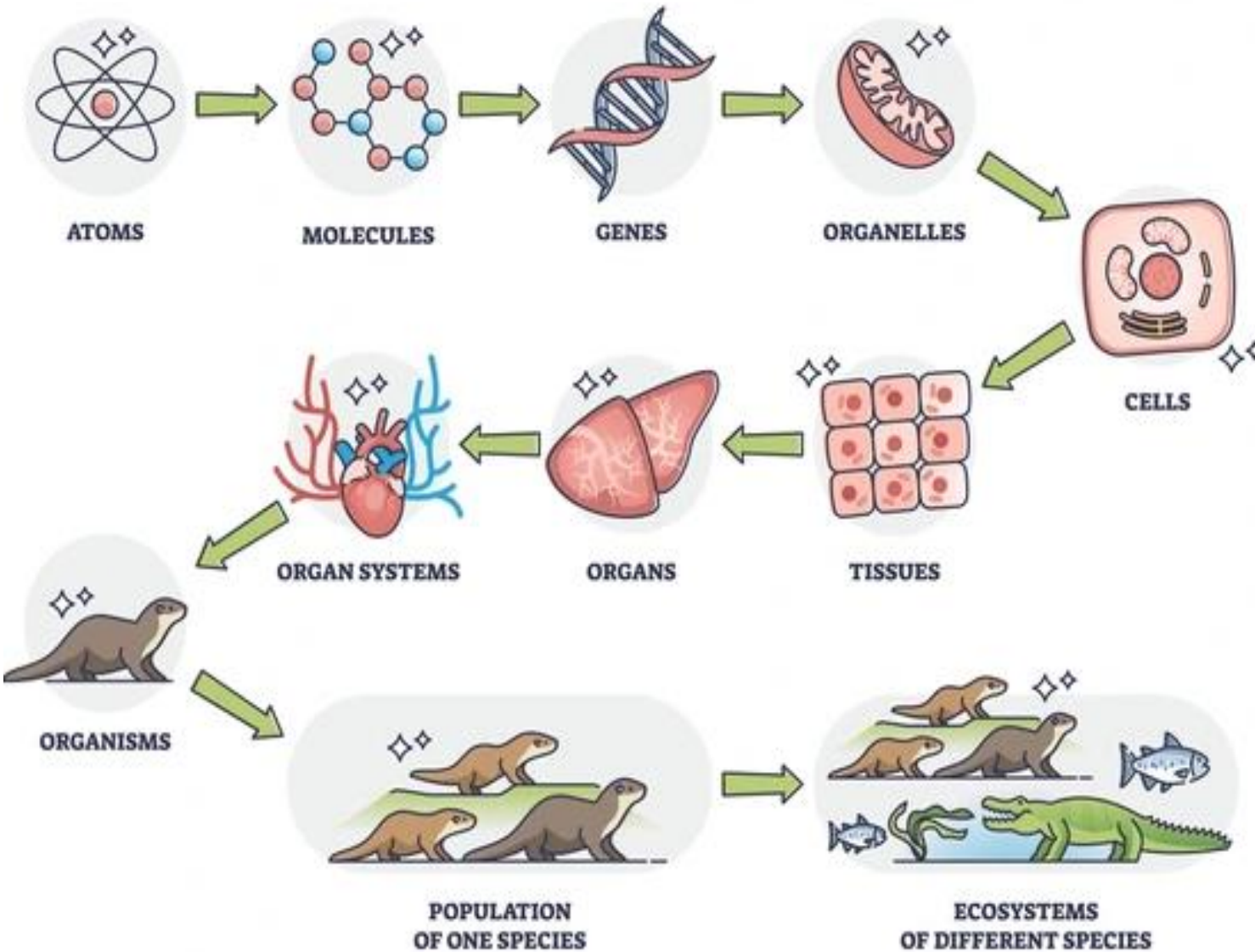
- We all are biological entities
- Know yourself, know how you function
- Understand your surroundings
- Do yourself a favor, save the world
- Research and Development in Upcoming Biology, Biotechnology

Beginning of Life: Embryonic Development

- Life Begets Life: Reproduction and Genetics
- You began life as a single cell (zygote/fusion)
- Your genes are found in your DNA, which is in turn found in your chromosomes



Introduction: Why Biology in Engineering



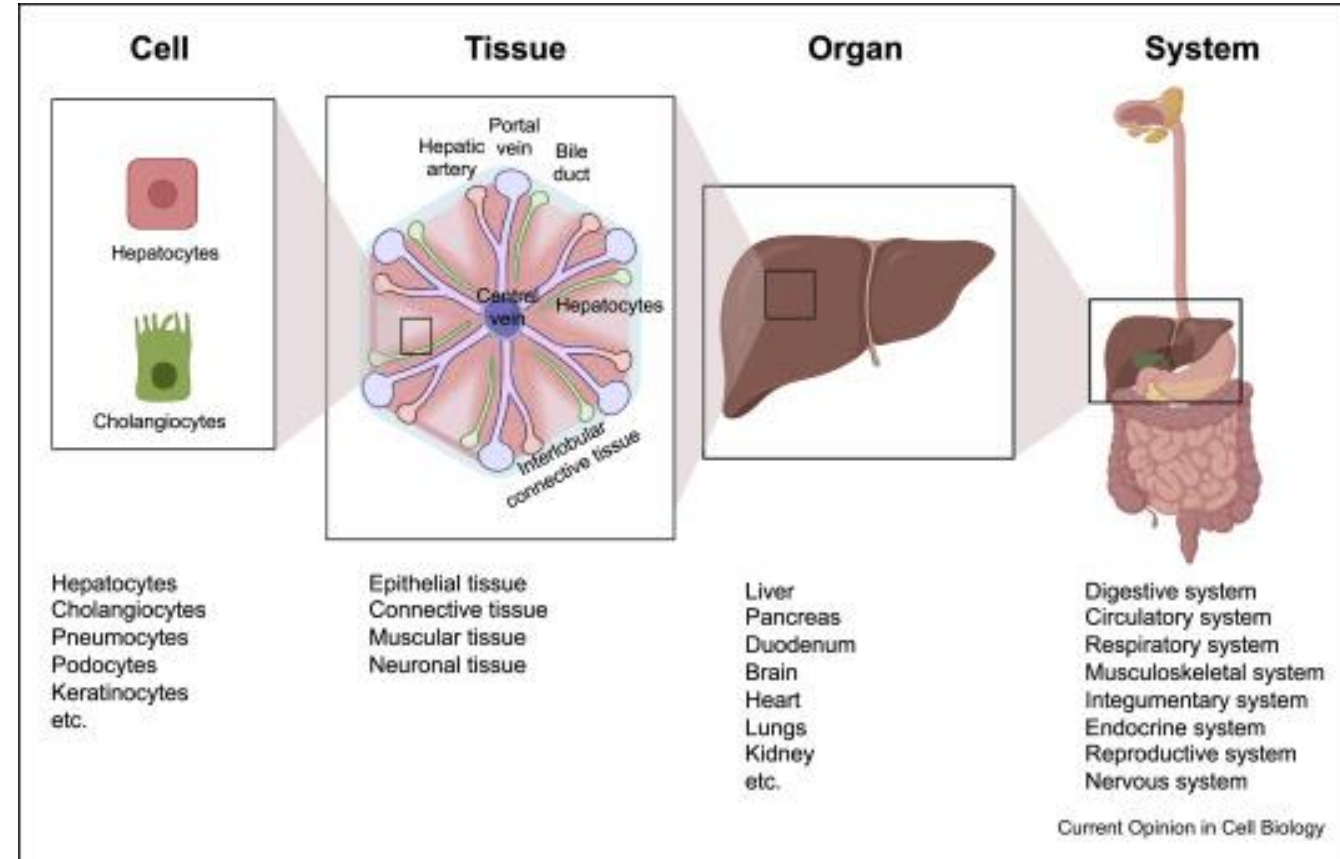
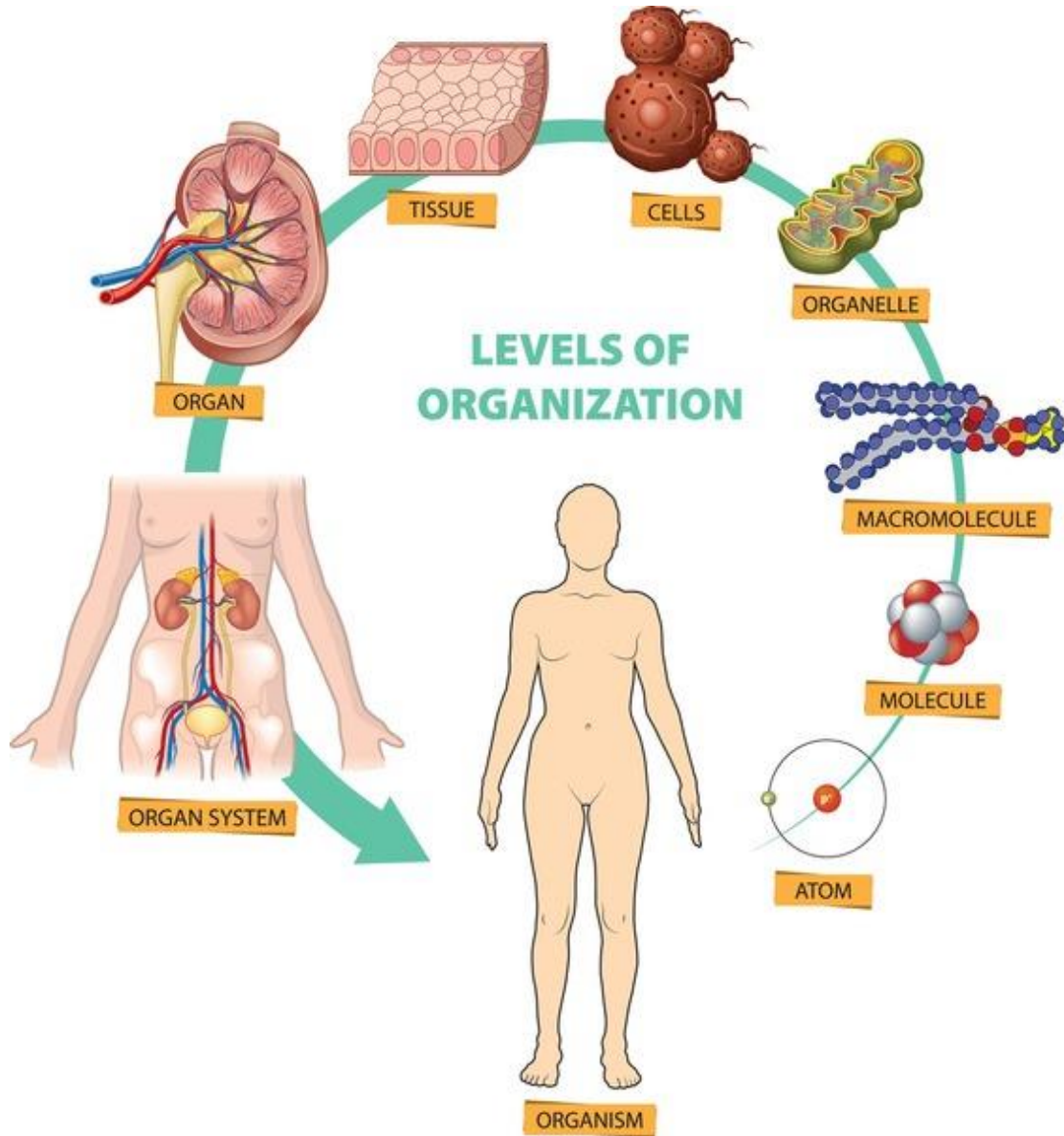
- Exploring the Living World
- It All Starts with a Cell: The absolute smallest unit of life is a single cell

Biocenosis:

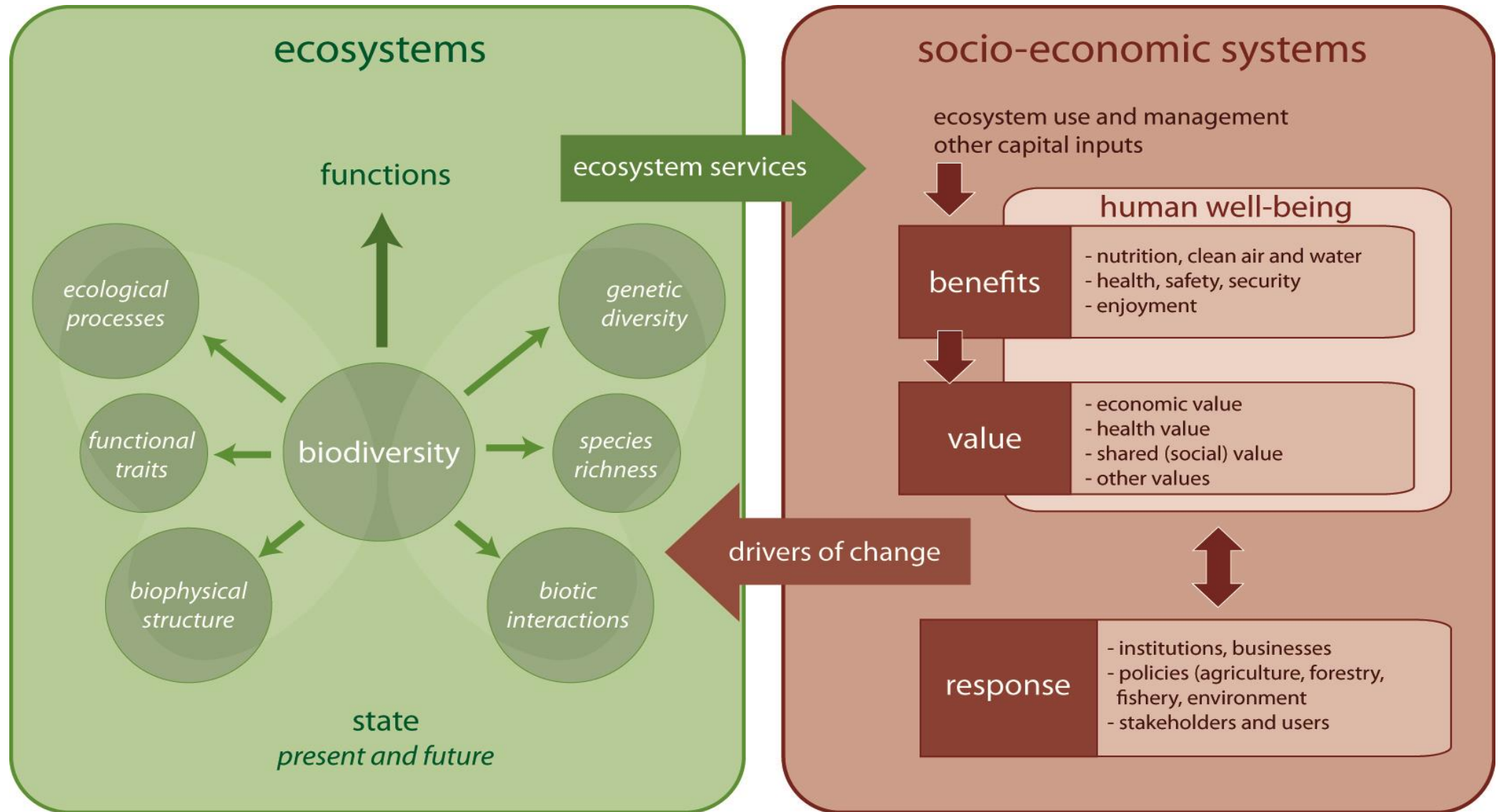
1. An association of different organisms forming a closely integrated community
2. The biosphere is a global ecosystem made up of living organisms (biota) and the nonliving (abiotic) factors that provide them with energy and nutrients..

Ecosystem: the interaction between organisms and the surrounding environment.

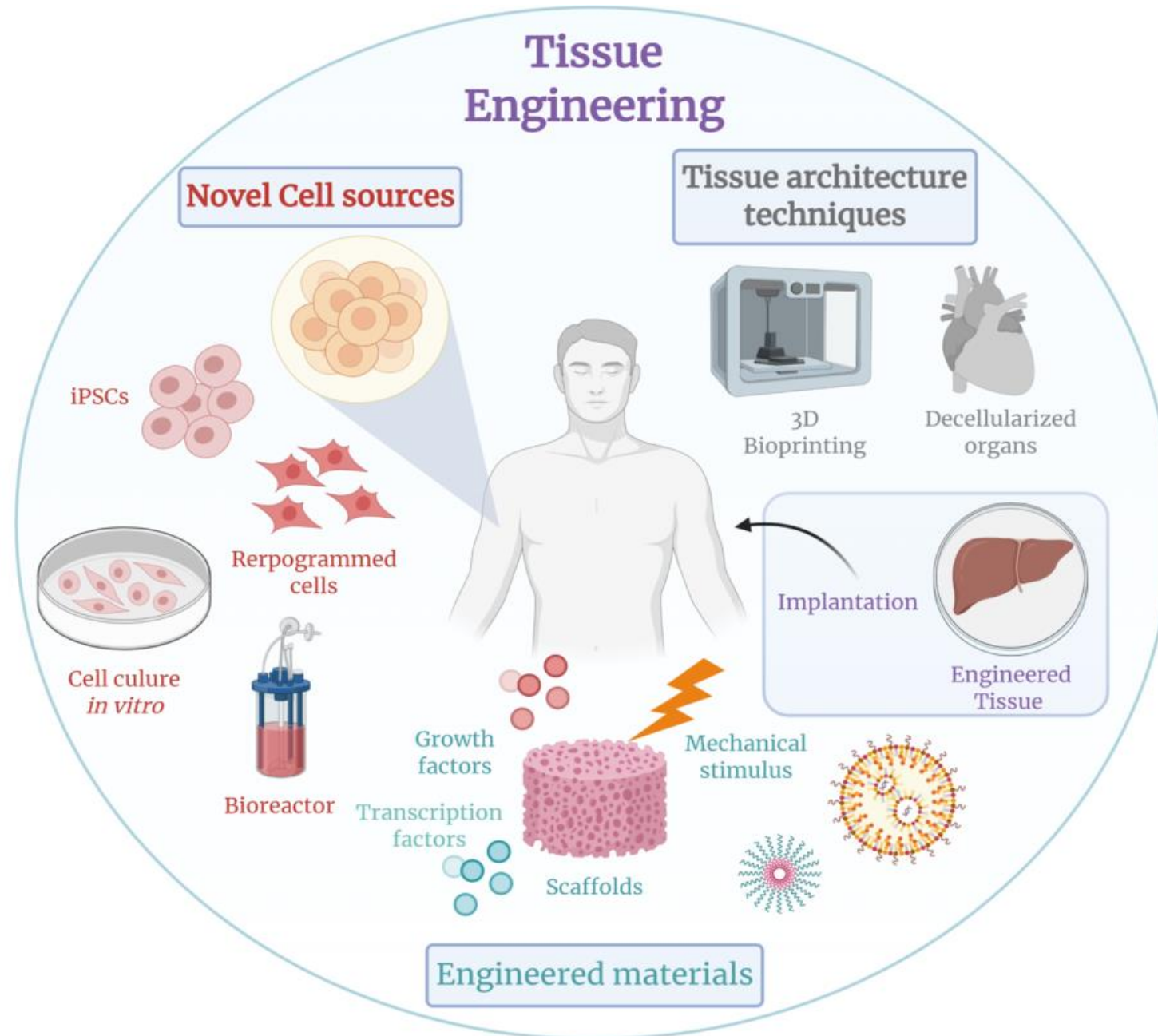
Organogenesis



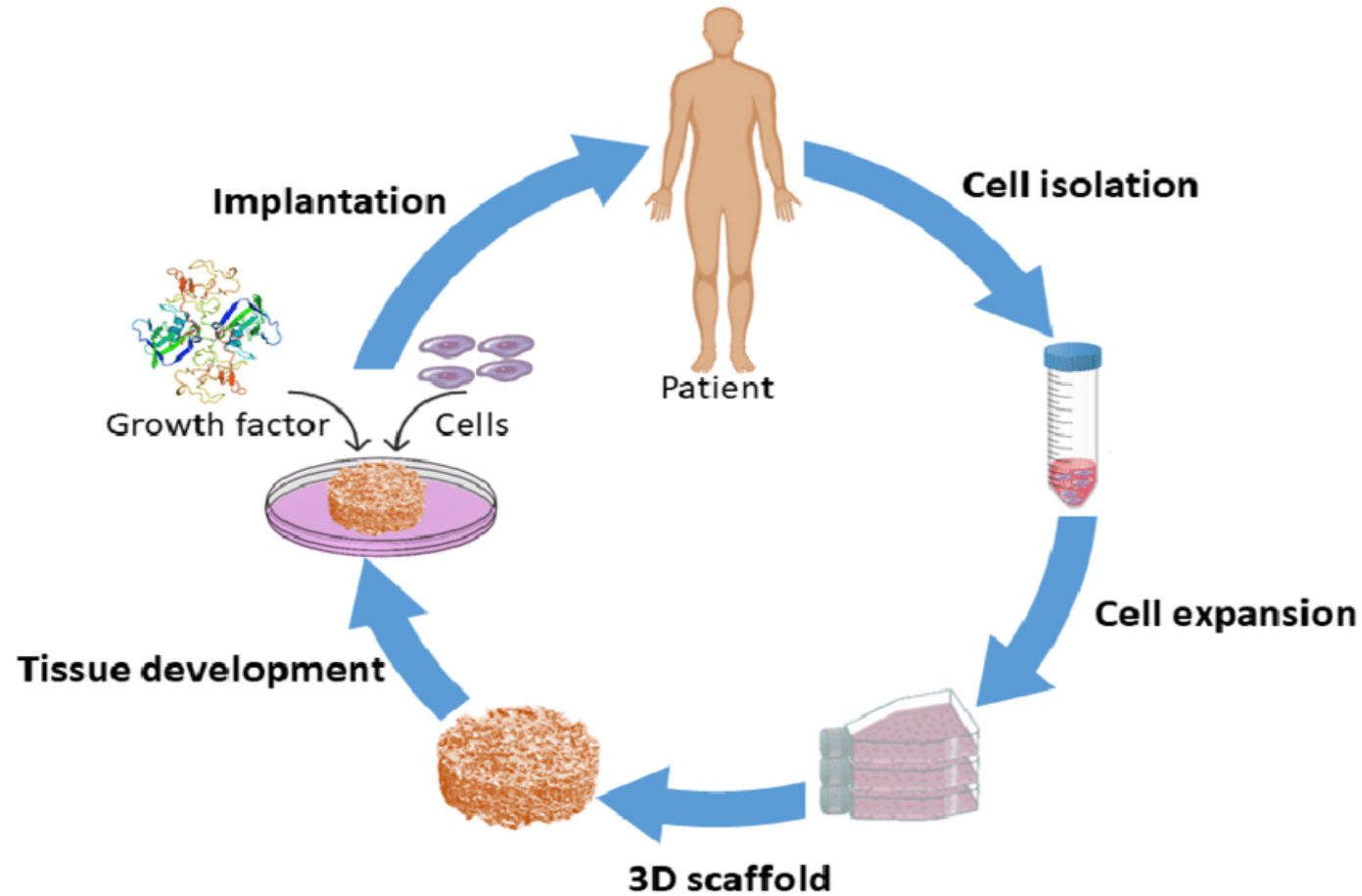
Why Biology in Engineering?



Introduction: Scope for Engineers



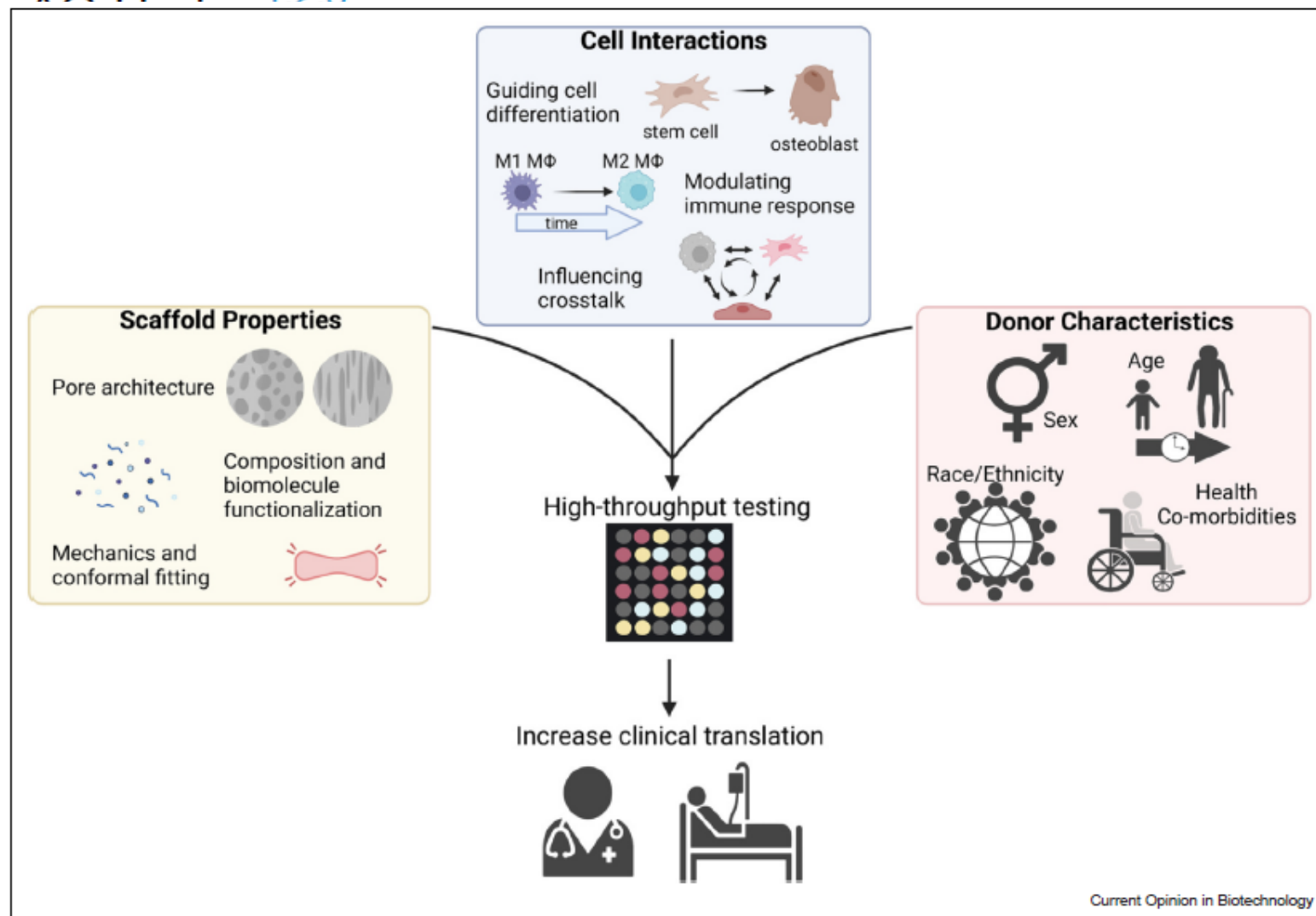
Introduction: Scope for Engineers



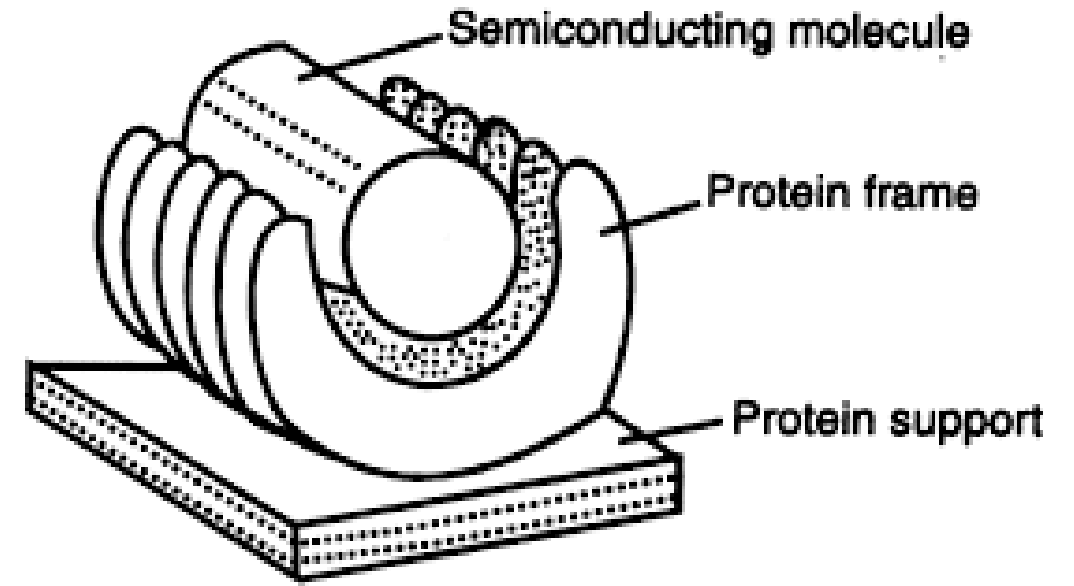
Muscular Bio-polymers These refer to specialized biopolymer nano-composites that can be used for artificial muscles, and so on. These specialized materials aim to have properties that are close to biological muscles such as excitability, contractility, elasticity, and extensibility. This example is only one advantage of engineers knowing biology, that is, inspiration or help with solutions to challenging problems.

Mineralized collagen scaffolds for regenerative engineering applications

Vasiliki Kolliopoulos¹ and Brendar A. G. et al.^{1,2,3}

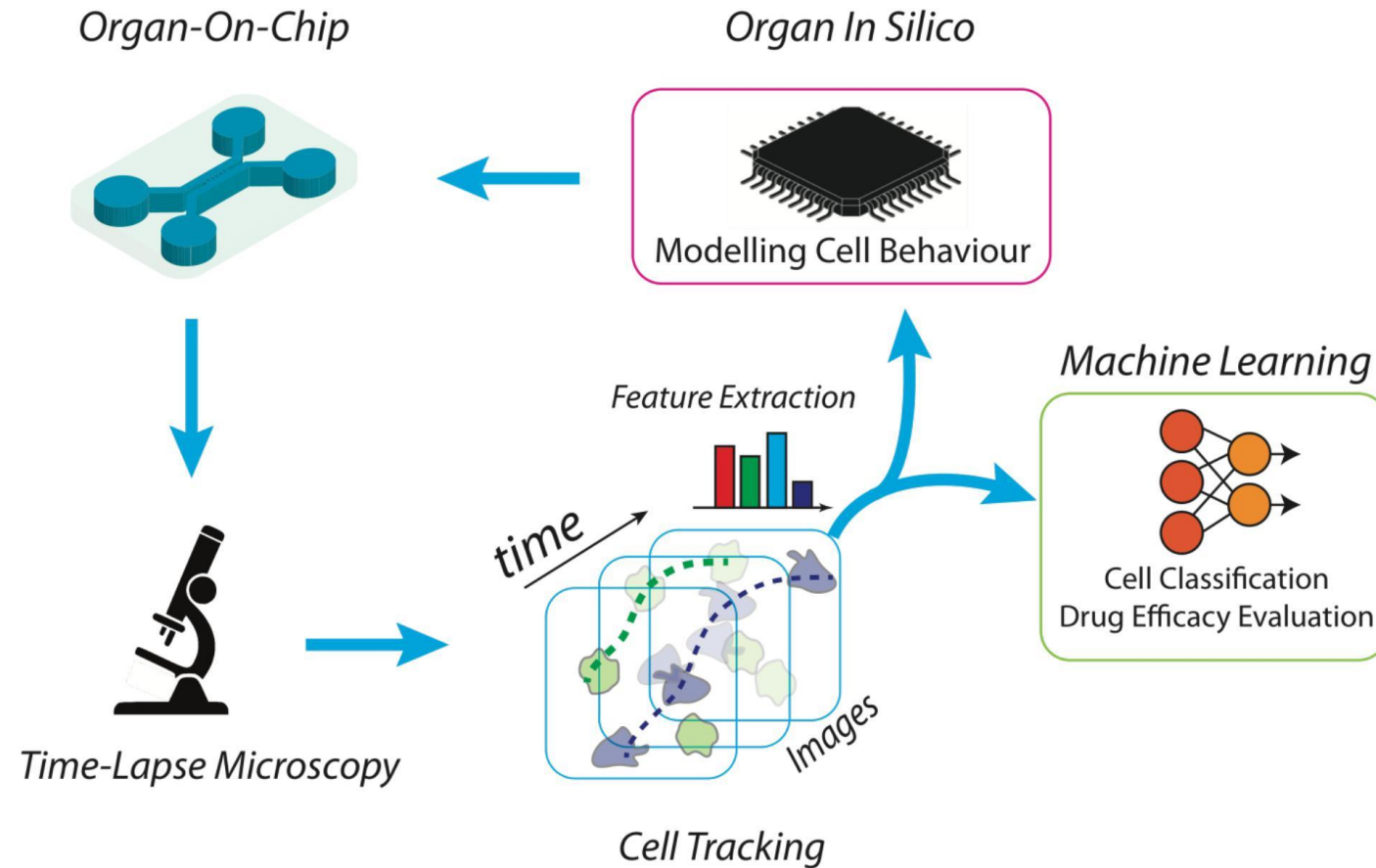


Introduction: Bio-chips



Bio-chips These are miniaturised laboratories in which thousands of biochemical reactions can be carried out simultaneously at micron scales for useful purposes such as disease studies, drug discovery and safety or toxicity studies.

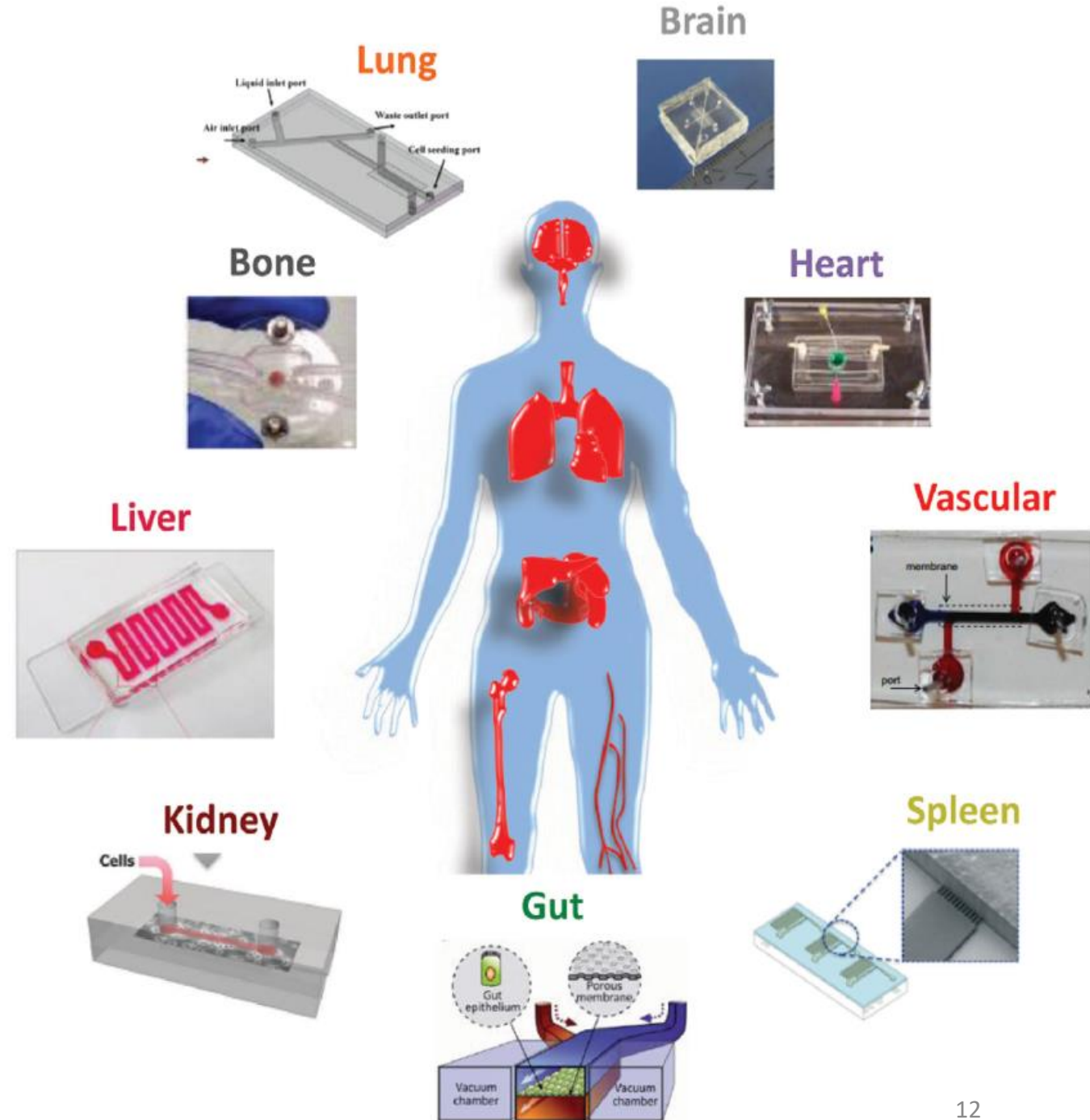
(mini/micro) Organ In a Chip



Organ-on-a-chip It is a micro set-up to test the effect of say, new drugs or toxins on representative animal tissues without actual animal studies. Microfluidic devices with small membranes and other support structures that can hold specialized organ cells are used to create the organs- on-a-chip. Body fluid-like materials that can flow on either side of the membrane, provide a functional representation of the real organ.

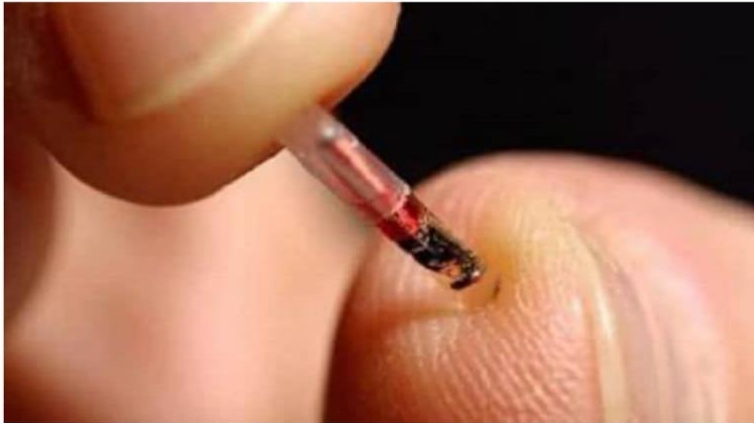
Organ-on-a-chip

Organ-on-a-chip Body fluid-like materials that can flow on either side of the membrane, provide a functional representation of the real organ. Drugs to be tested are dissolved in those body fluid-like materials that are contacted with the tissue to test their effects on the organ tissues. The claim is that these organs-on-a-chip are closer to the human situation than the animals are, and thus the predictions could be much better. Further, whole *organs* can potentially be *3-D bio-printed* for various study purposes. However, 3-D bio-printing is a challenge because the biological cells need to be kept alive and vibrant during processing and printing. Cells need suitable nutrition and environment to be active, and techniques are being developed to maintain cells under desirable conditions.



Global Biochips Market Is Thriving Globally With Microsoft, Intel Corporation, Google, Inc., Apple, Inc., Qualcomm Technologies, Inc., Infineon Technologies AG, Microchip Technology, Inc., Omnivision Technologies, Inc., Sony Corporation, Eyesight Technologies

01-18-2019 12:34 AM CET | IT, New Media & Software
Press release from: [Data Bridge Market Research](#) / PR
Agency: Data Bridge Market Research



Market Type:

- DNA Chip
- Lab-on-a-chip
- Protein Chips
- Microarrays
- Microfluidics

The major players in global Biochips market include:

- Abbott
- F. Hoffmann-La Roche Ltd.
- Thermo Fisher Scientific
- Cepheid
- Bio-Rad Laboratories, Inc.
- General Electric Company
- Illumina, Inc.
- FLUIDIGM
- PerkinElmer Inc.
- Agilent Technologies
- Predictive Biosciences, Inc.
- MYcroarray, Digital Genomics, Inc
- Elim Biopharmaceuticals, Inc.
- CustomArray Inc.
- Falcon Genomics, Inc., Genalyte, Inc.
- WaferGen Bio-systems, Inc.

End user:

- Hospital
- Diagnostic Centers
- Biotechnology & Pharmaceutical Companies
- Academic & Research Institutes

Scopes and opportunities for CSE Graduates in:

- **Academia & Research** – Universities, research institutes, and government organizations.
- **Biotech & Pharma Companies** – Pfizer, Moderna, Roche, Illumina, Thermo Fisher Scientific, etc.
- **Healthcare & Hospitals** – AI-driven diagnostics, medical informatics, and healthcare analytics.
- **Tech & AI Companies** – Google DeepMind, IBM Watson, Microsoft AI for Health, NVIDIA BioNeMo.
- **Startups & Entrepreneurship** – AI-driven biotech, personalized medicine, and bioinformatics startups.

Scopes and opportunities for CSE Graduates in:

1. Bioinformatics and Computational Biology:

- Developing algorithms and software for analyzing biological data, such as DNA sequencing, protein structures, and gene expression.
- Working with large datasets (Big Data) for genomics, proteomics, and transcriptomics.
- Conducting computational modeling and simulations of biological processes.

2. Artificial Intelligence and Machine Learning in Biology:

- Developing AI-driven tools for disease diagnosis, drug discovery, and personalized medicine.
- Using machine learning to predict protein structures, analyze medical images, and detect genetic disorders.
- Enhancing medical research with deep learning applications.

3. Biotechnology and Biomedical Engineering:

- Contributing to the development of biotechnological solutions, such as CRISPR gene editing and synthetic biology.
- Designing software for laboratory automation and robotic systems used in biotech research.
- Creating computational models for drug interactions and metabolic pathways.

Scopes and opportunities for CSE Graduates in:

4. Healthcare and Medical Informatics:

- Developing electronic health records (EHR) systems, health data analytics, and predictive modeling.
- Designing AI-powered diagnostic tools for early disease detection.
- Improving patient care through personalized medicine approaches.

5. Systems Biology and Mathematical Modeling:

- Using computational methods to model complex biological systems.
- Simulating disease progression, immune responses, and metabolic networks.
- Assisting in agricultural and environmental biotechnology through ecosystem modeling.

6. Pharmaceutical and Drug Discovery:

Applying AI and machine learning to accelerate drug discovery and development.

- Conducting computational drug design to identify potential drug candidates.
- Developing software for clinical trial analysis and optimization.

7. Biomedical Image Processing:

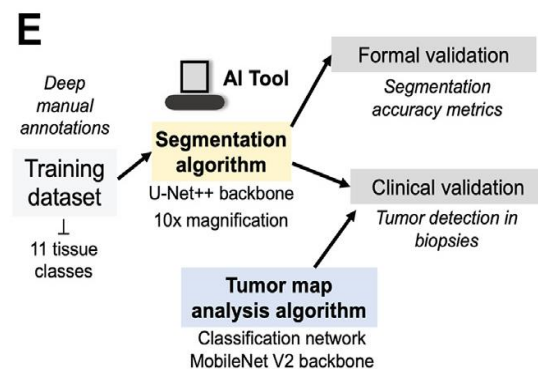
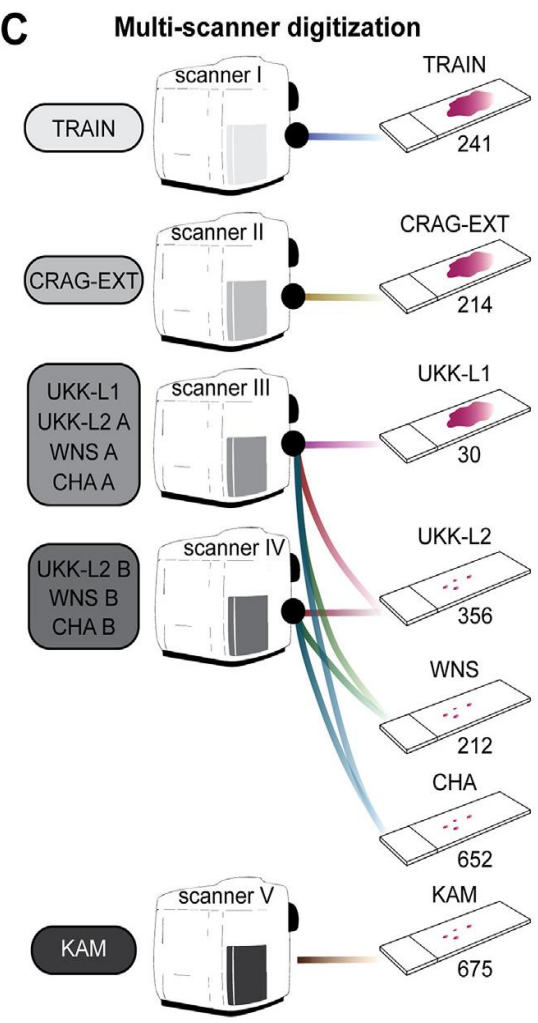
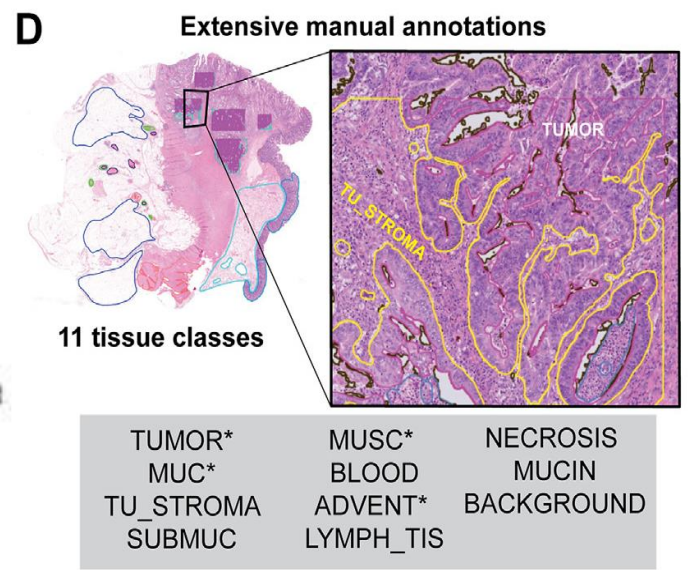
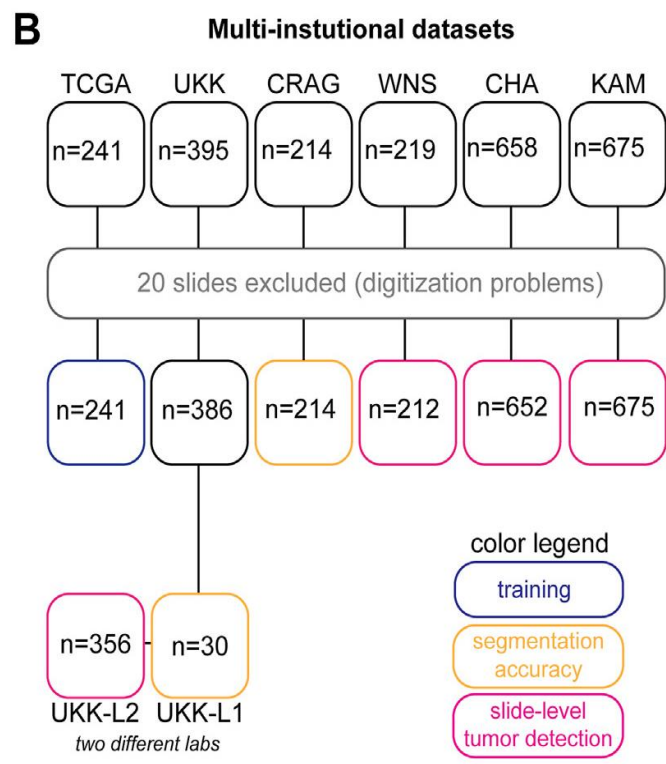
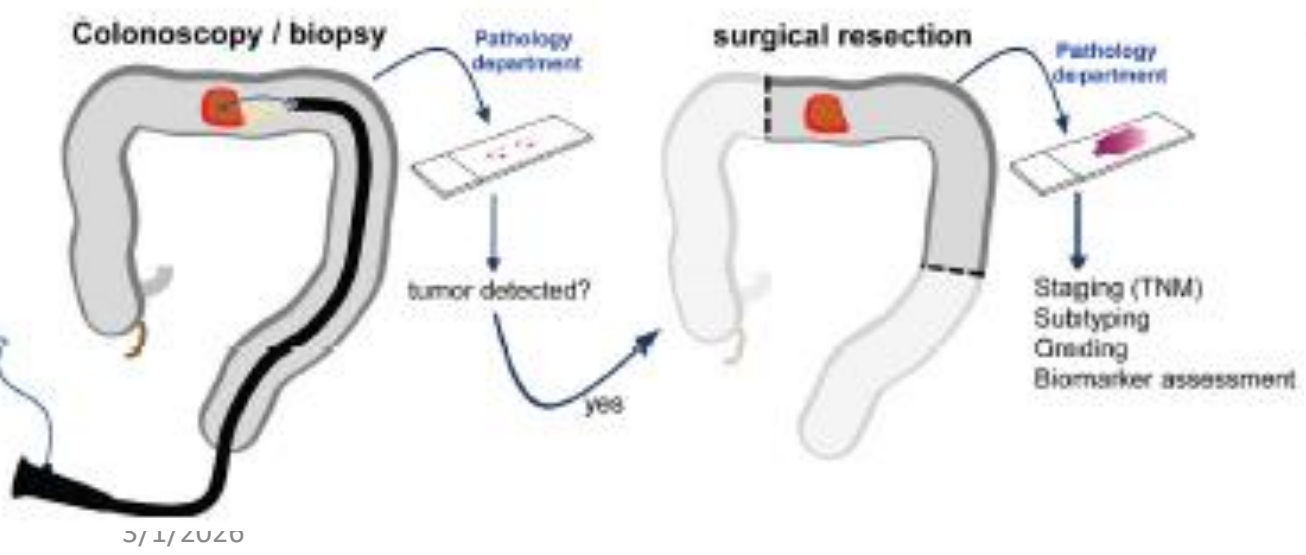
- Enhancing medical imaging techniques such as MRI, CT scans, and X-rays using AI.
- Developing image analysis tools for detecting diseases like cancer.
- Using deep learning for automated analysis of histopathological images.

Research Article

Artificial Intelligence–Based Tool for Tumor Detection and Quantitative Tissue Analysis in Colorectal Specimens

Johanna Griem^a, Marie-Lisa Eich^a, Simon Schallenberg^b, Alexey Pryalukhin^c, Andrey Bychkov^{d,e}, Junya Fukuoka^{d,e}, Vitaliy Zayats^f, Wolfgang Hulla^c, Jijee Munkhdelger^e, Alexander Seper^g, Tsvetan Tsvetkov^a, Anirban Mukhopadhyay^h, Antoine Sanner^h, Jonathan Stieber^h, Moritz Fuchs^h, Niklas Babendererde^h, Birgid Schömig-Markieffka^a, Sebastian Klein^a, Reinhard Buettner^a, Alexander Quaas^a, Yuri Tolkach^{a,*}

^aInstitute of Pathology, University Hospital Cologne, Cologne, Germany; ^bInstitute of Pathology, University Hospital Charité, Berlin, Germany; ^cInstitute of Pathology, State Hospital Wiener Neustadt, Wiener Neustadt, Austria; ^dDepartment of Pathology Informatics, Nagasaki University Graduate School of Biomedical Sciences, Nagasaki, Japan; ^eDepartment of Pathology, Kameda Medical Center, Kamogawa, Japan; ^fLaboratory for Medical Artificial Intelligence, The Resource Center for Universal Design and Rehabilitation Technologies (RCUD and RT), Moscow, Russia; ^gDanube Private University, Medical Faculty, Krems-Stein, Austria; ^hTechnical University Darmstadt, Darmstadt, Germany



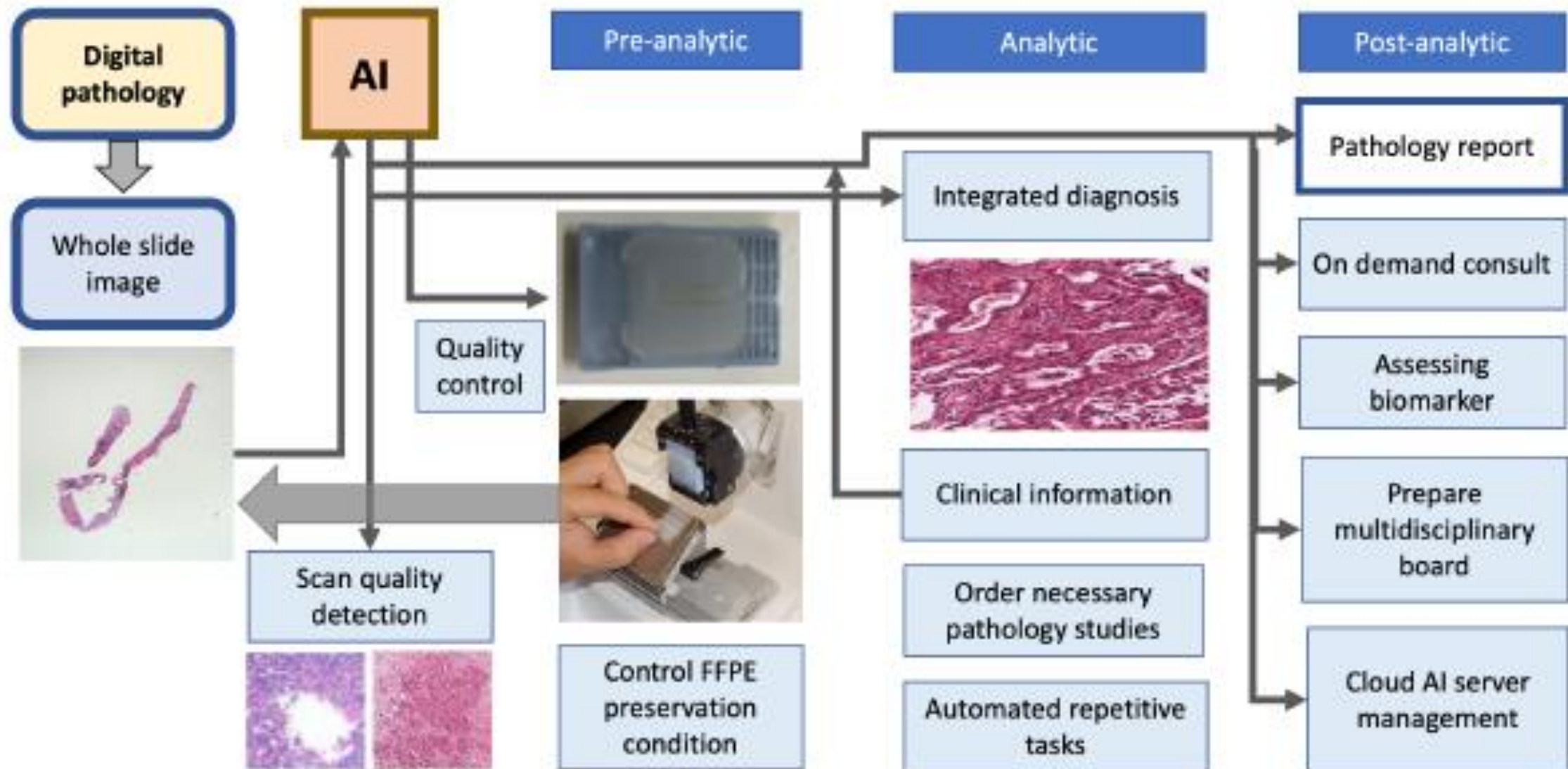
Artificial Intelligence in Pathology

Sebastian Försch, Frederick Klauschen, Peter Hufnagl, Wilfried Roth

TABLE 1

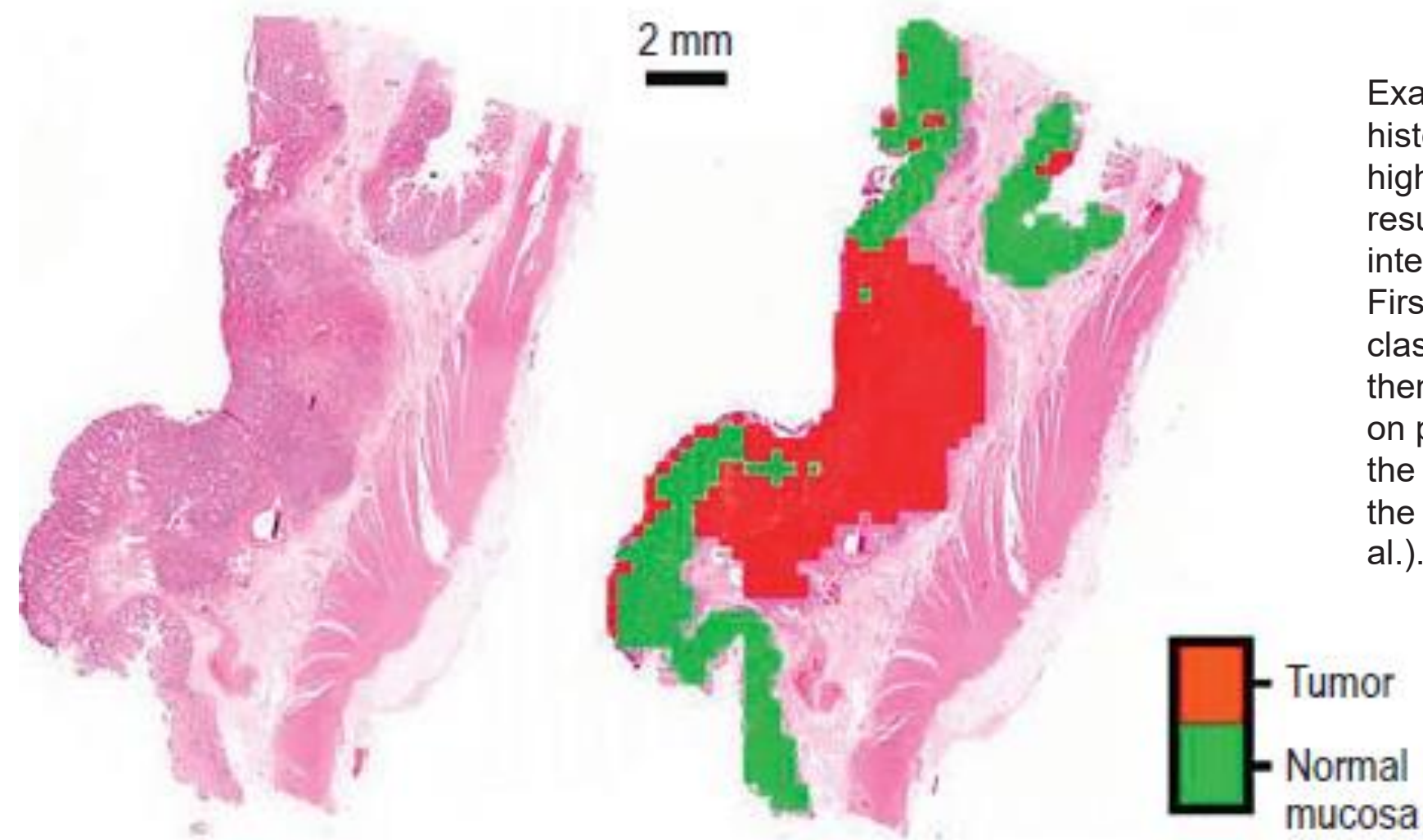
Overview of key studies

Reference	Entity	Sample size	Method	Type of study	Result
Ehteshami Bejnordi et al. 2017 (14)	Breast cancer	270	Variety of deep learning algorithms	Comparative study of retrospective cases	The best model (AUROC: 0.960) was superior in diagnostic accuracy to eleven pathologists if these had to work under time pressure (AUROC: 0.810).
Mukhopadhyay et al. 2018 (3)	Various	1992	Digitalized tissue slides	Randomized, blinded non-inferiority trial	Digital interpretation is not inferior to microscopy-based interpretation.
Steiner et al. 2018 (15)	Breast cancer	70	Deep convolutional network	Comparative study of retrospective cases	Pathologists achieved higher sensitivity with AI support (increase from 83.3% to 91.2%).
Campanella et al. 2019 (17)	Breast cancer Prostate cancer Basal cell carcinoma	15 187	Multiple instance learning	Retrospective study	Without the need for manual tumor annotation, the system achieved AUROCs of up to 0.991 and could independently classify 65–75% of tissue specimens without loss in sensitivity.



Embedding AI into pathology department workflow. The digital pathology supplies whole-slide images to artificial intelligence, which performs quality control of pre-analytic phase, analytic phase and post-analytic phase of pathology laboratory process.

Deep learning model can differentiate colorectal cancer from normal colorectal mucosa.



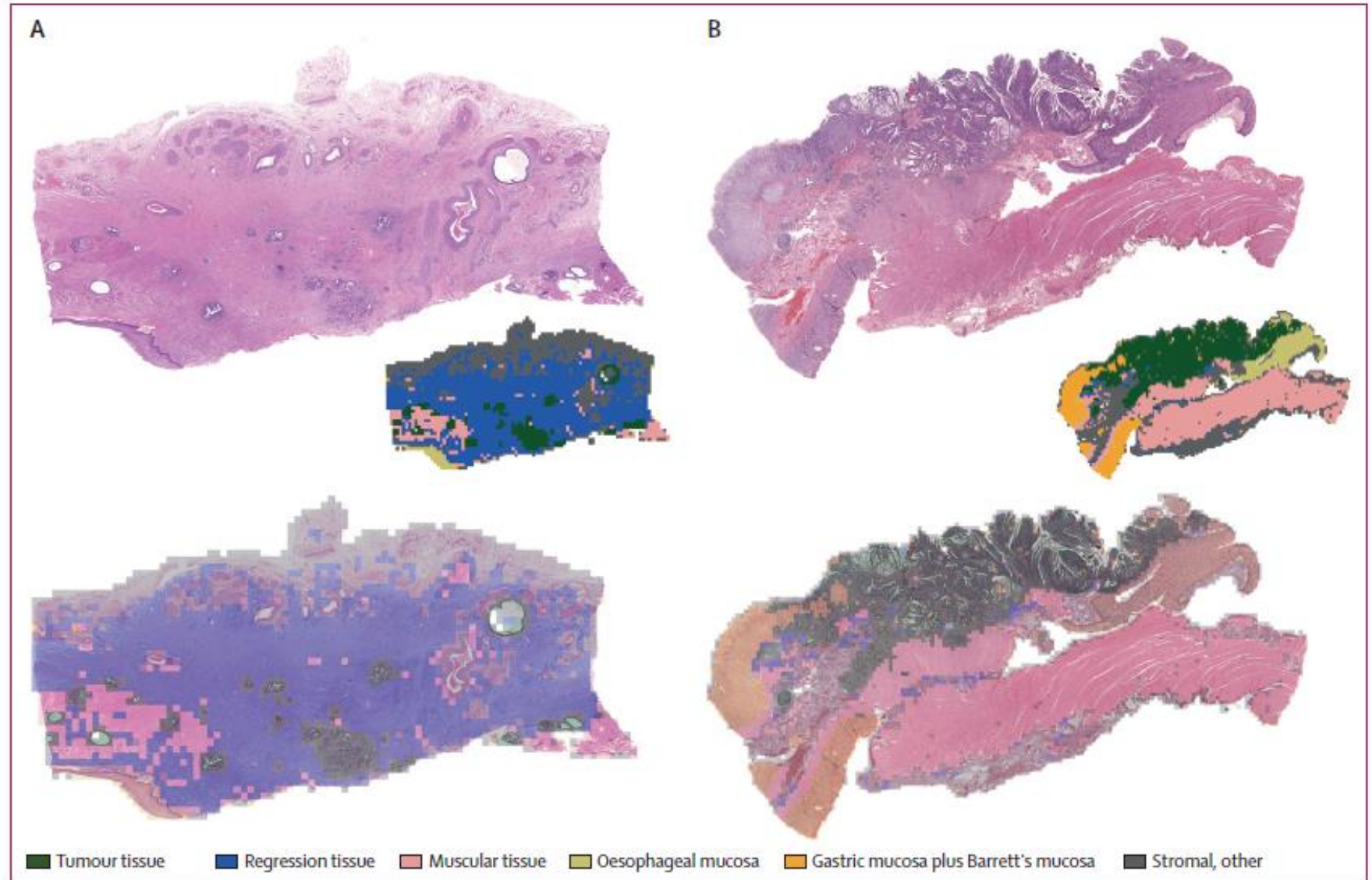
Example of On the left, the conventional histological input image; on the right, highlighting of the tissue according to the result of classification by the artificial intelligence model.

First, individual image sections (tiles) are classified by the artificial neural network and then each individual tile is color-coded based on prediction probability: higher probability of the class “tumor”: red; higher probability of the class “normal mucosa”: green (Försch et al.).

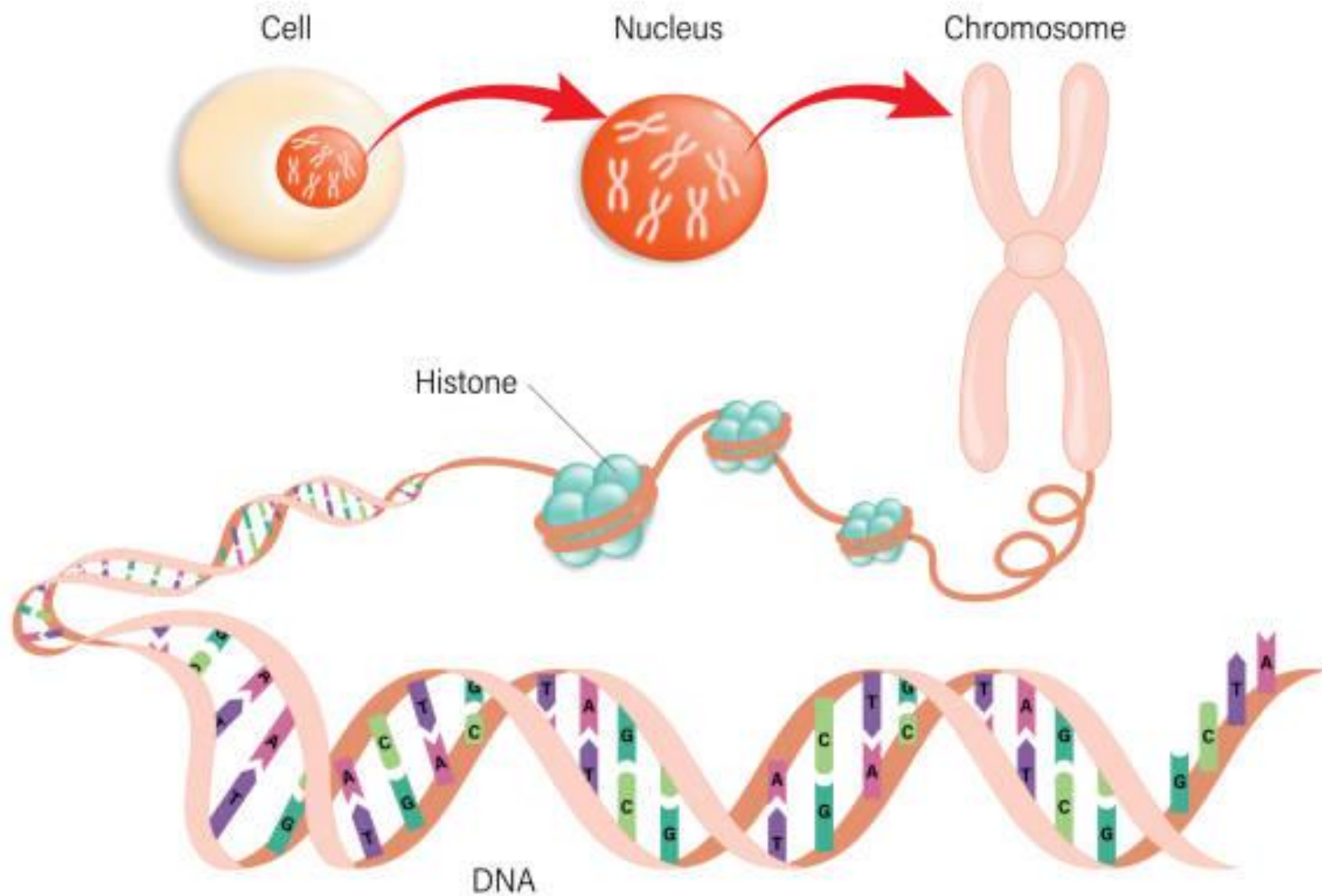
Artificial intelligence for tumour tissue detection and histological regression grading in oesophageal adenocarcinomas: a retrospective algorithm development and validation study

Examples from the Wiener Neustadt cohort (A) and The Cancer Genome Atlas cohort (B) of whole-slide inference by the artificial intelligence tool. The tool was using the final version of the algorithm.

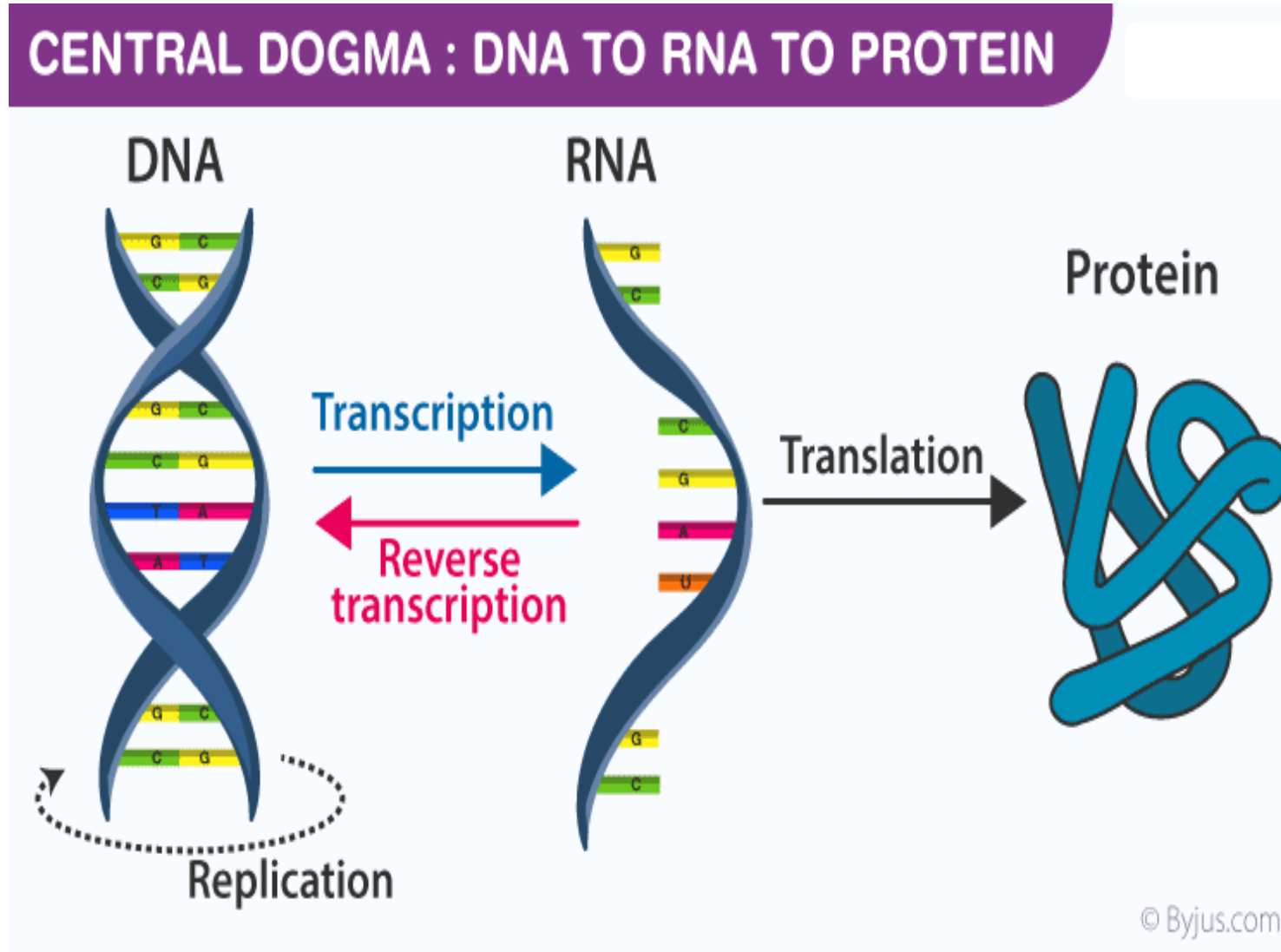
Lancet Digit Health 2023;
5: e265–75



Cell, Nucleus, Chromosome, Histone and DNA

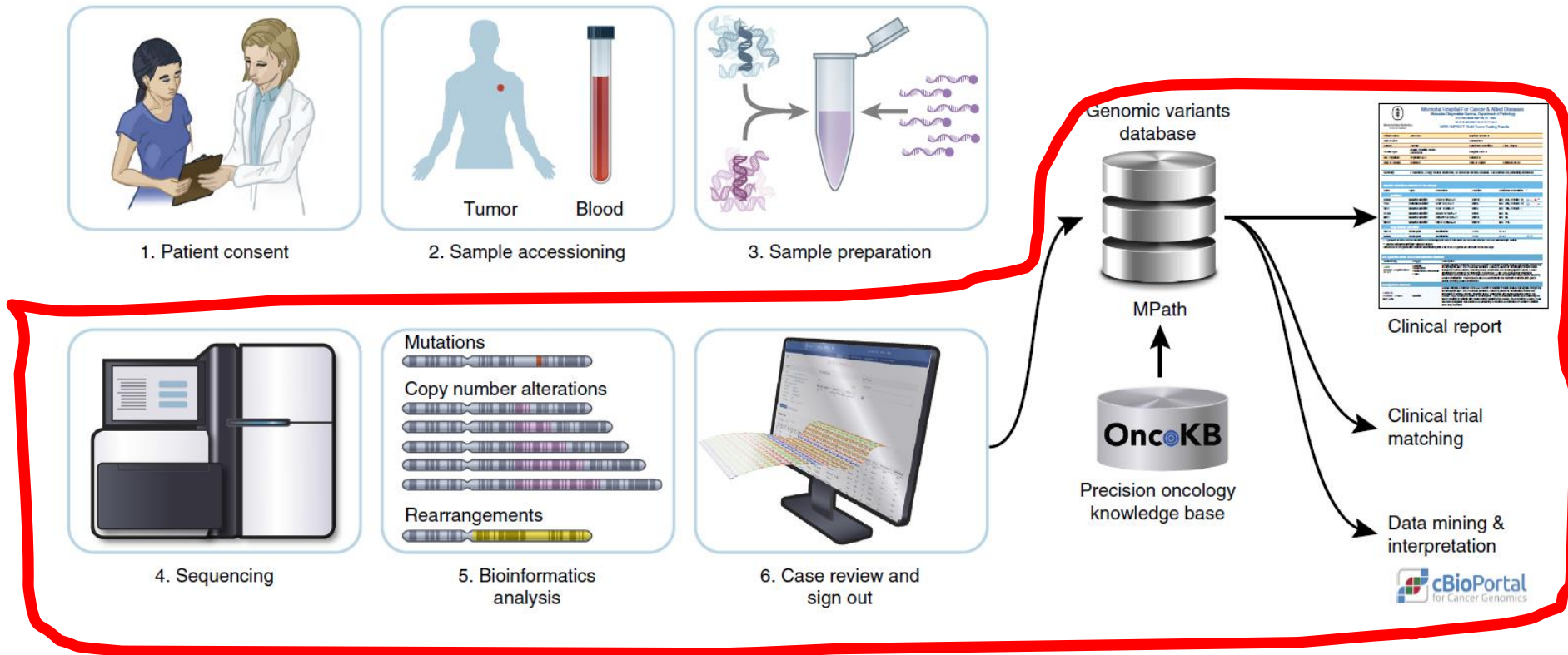


Mechanism of gene transcription and translation.



Do CSE students really need to know this?

Schematic diagram for the mutations discovery from Hepatocellular carcinoma (HCC) patients:



Mutational landscape of metastatic cancer revealed from prospective clinical sequencing of 10,000 patients

<https://www.nature.com/articles/nm.4333>

[Nature Medicine](#) volume 23, pages703–713 (2017)

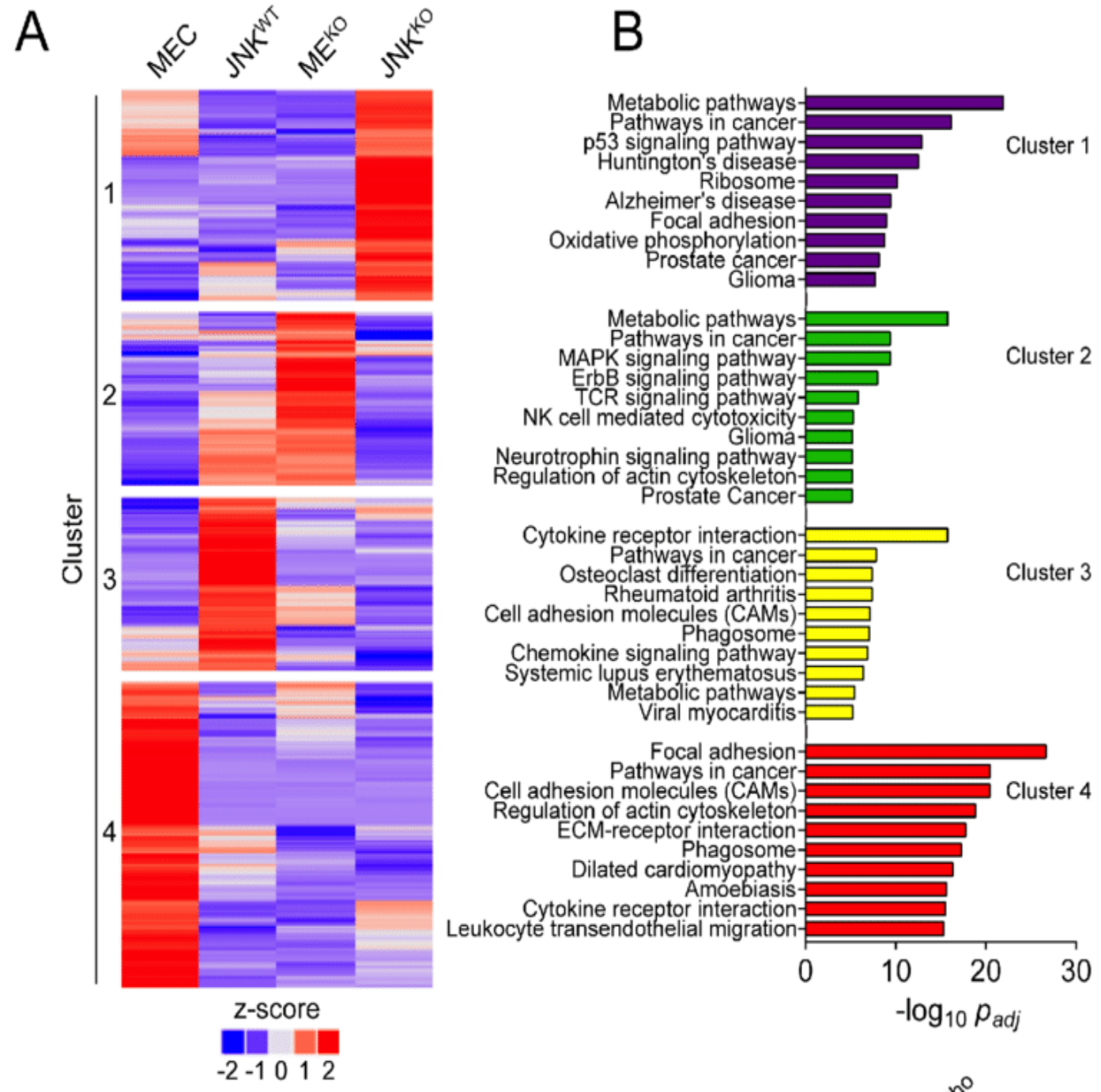
Why we need to know these stuff !!??

The cJUN NH2-terminal kinase (JNK) signaling pathway promotes genome stability and prevents tumor initiation.

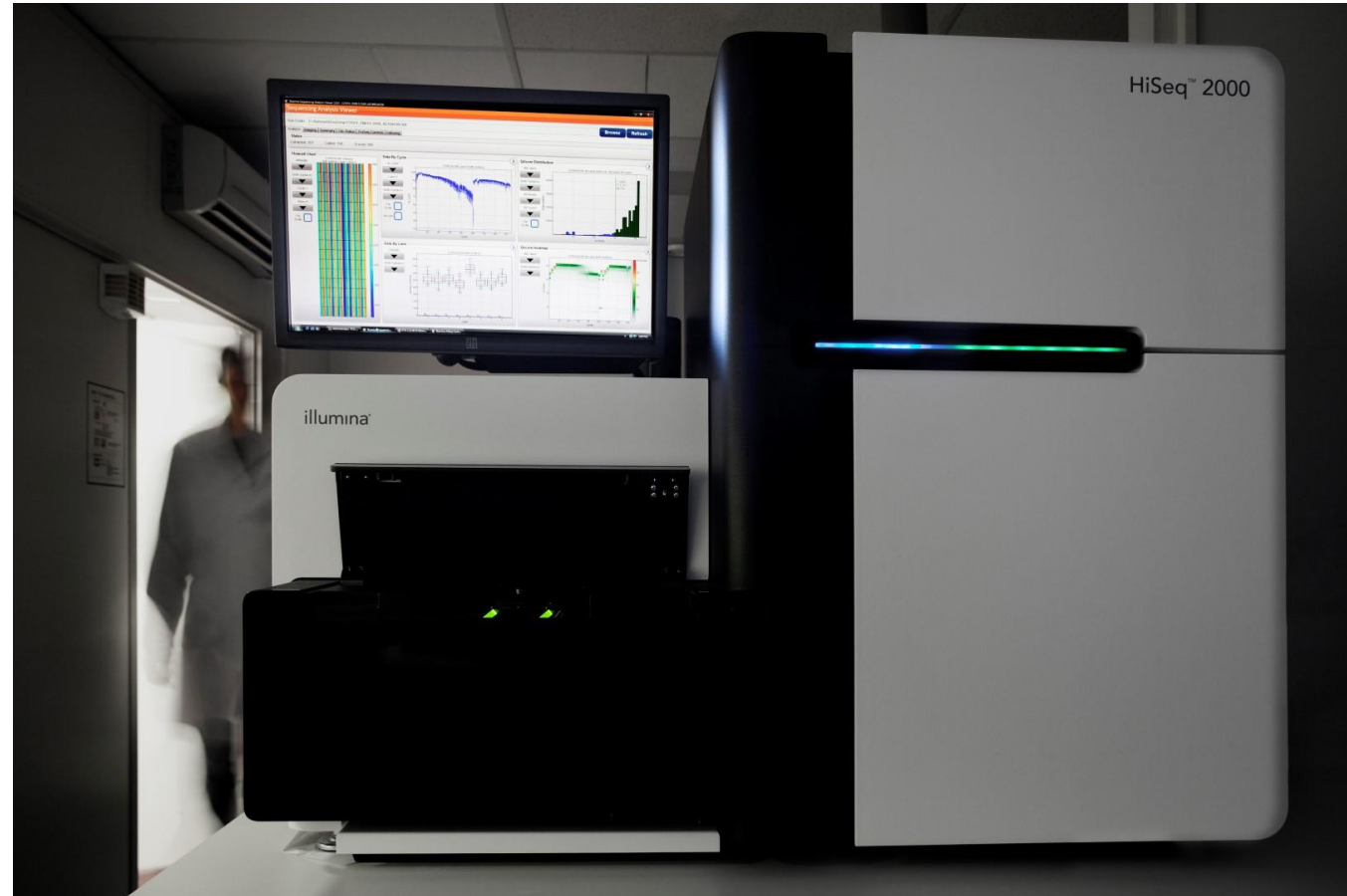
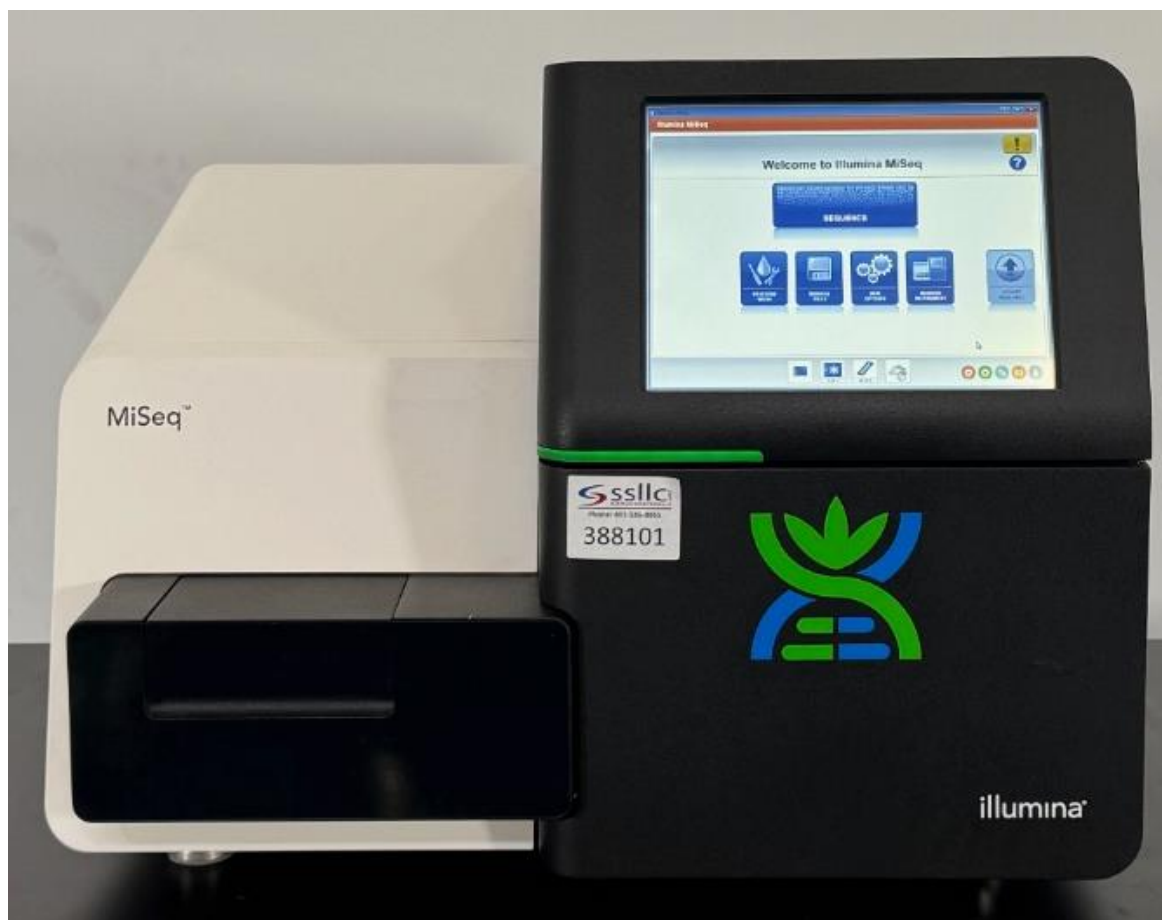
University of Massachusetts Medical School,
Worcester, United States.

<https://elifesciences.org/articles/36389>

<https://www.nature.com/articles/s41698-021-00142-x>



Next Generation Sequencing (NGS) to detect mutation in DNA



Identification of mutated protein/gene

Gene name	Wild Type	Mutant
MTOR	TWYEKLHEWEDALVAYDKKMDTNKD	TWYEKLHEWEDAPVAYDKKMDTNKD
TMCC2	PDEKERSPEMHRVSYAMSLHDLPAR	PDEKERSPEMHRISYAMSLHDLPAR
FAM71A	ASIPKTSTELAEEPATGGIKEAAAA	ASIPKTSTELAEKPATGGIKEAAAA
CDC42BPA-1	ETDPVENTYVWNPSVKFHIQSRSTS	ETDPVENTYVWNLSVKFHIQSRSTS
NID1	EGRVPQIPFGSSVHIEPYTELYHYS	EGRVPQIPFGSSMHIEPYTELYHYS
ZDHC18	HCSVCDNCVERFDHHCPWVGNCVGR	HCSVCDNCVERFNHHCPWVGNCVGR
OSBPL9	KPYNPILGEIFQCHWTLPNDTEENT	KPYNPILGEIFQFHWTLPLNDTEENT
CAMTA1	LVSGDSTALSQSEDGARAPFTQAEM	LVSGDSTALSQSQD GARAPFTQAEM

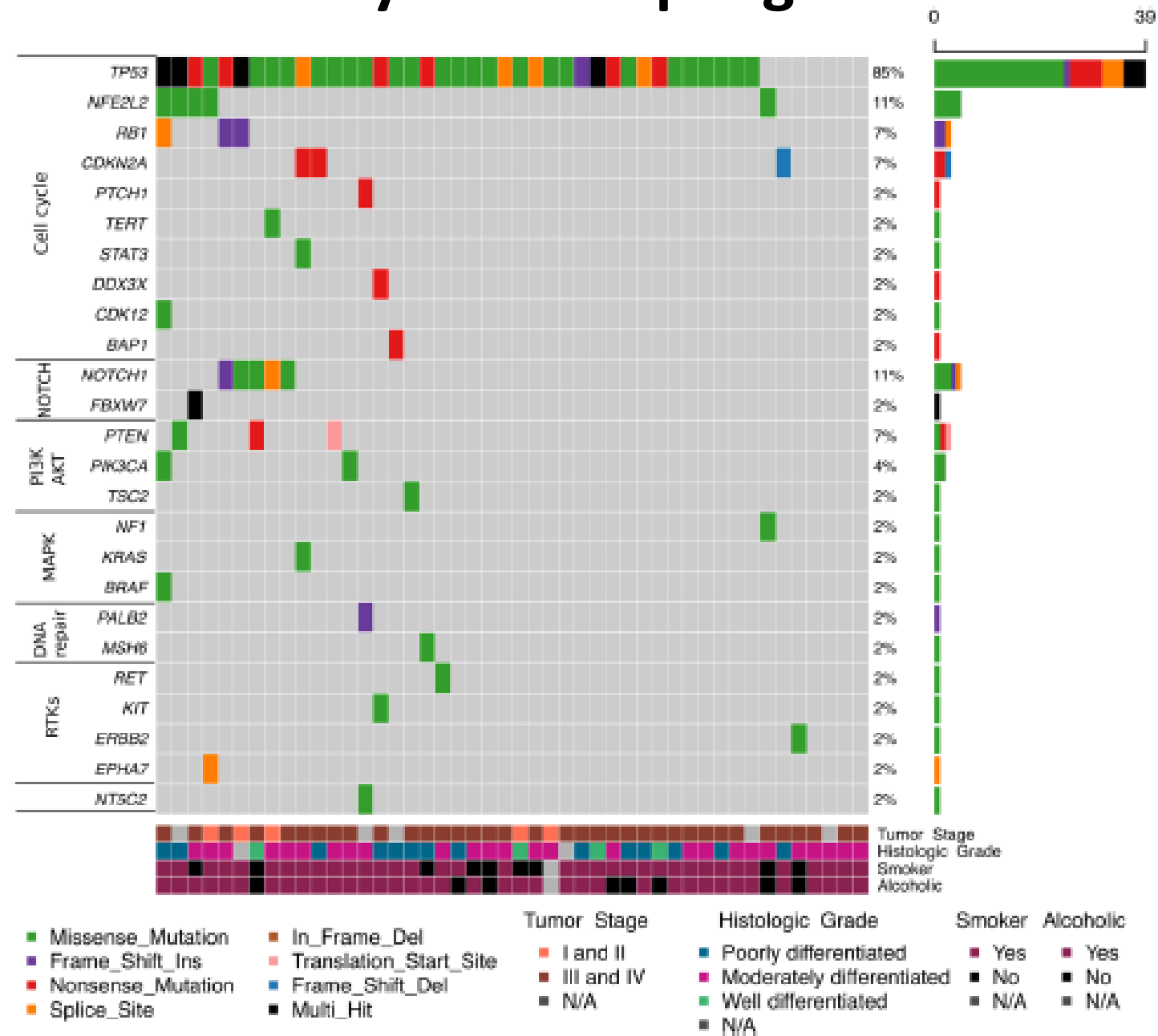
OPEN

Profile of esophageal squamous cell carcinoma mutations in Brazilian patients

Fernanda Franco Munari¹, Wellington dos Santos¹, Adriane Feijó Evangelista¹, Ana Carolina Carvalho¹, Paula Aguiar Pastrez¹, Diego Bugatti², Durval R. Wohnrath², Cristovam Scapulatempo-Neto^{1,3}, Denise Peixoto Guimarães^{1,4}, Adhemar Longatto-Filho^{1,5,6,7} & Rui Manuel Reis^{1,6,7}✉

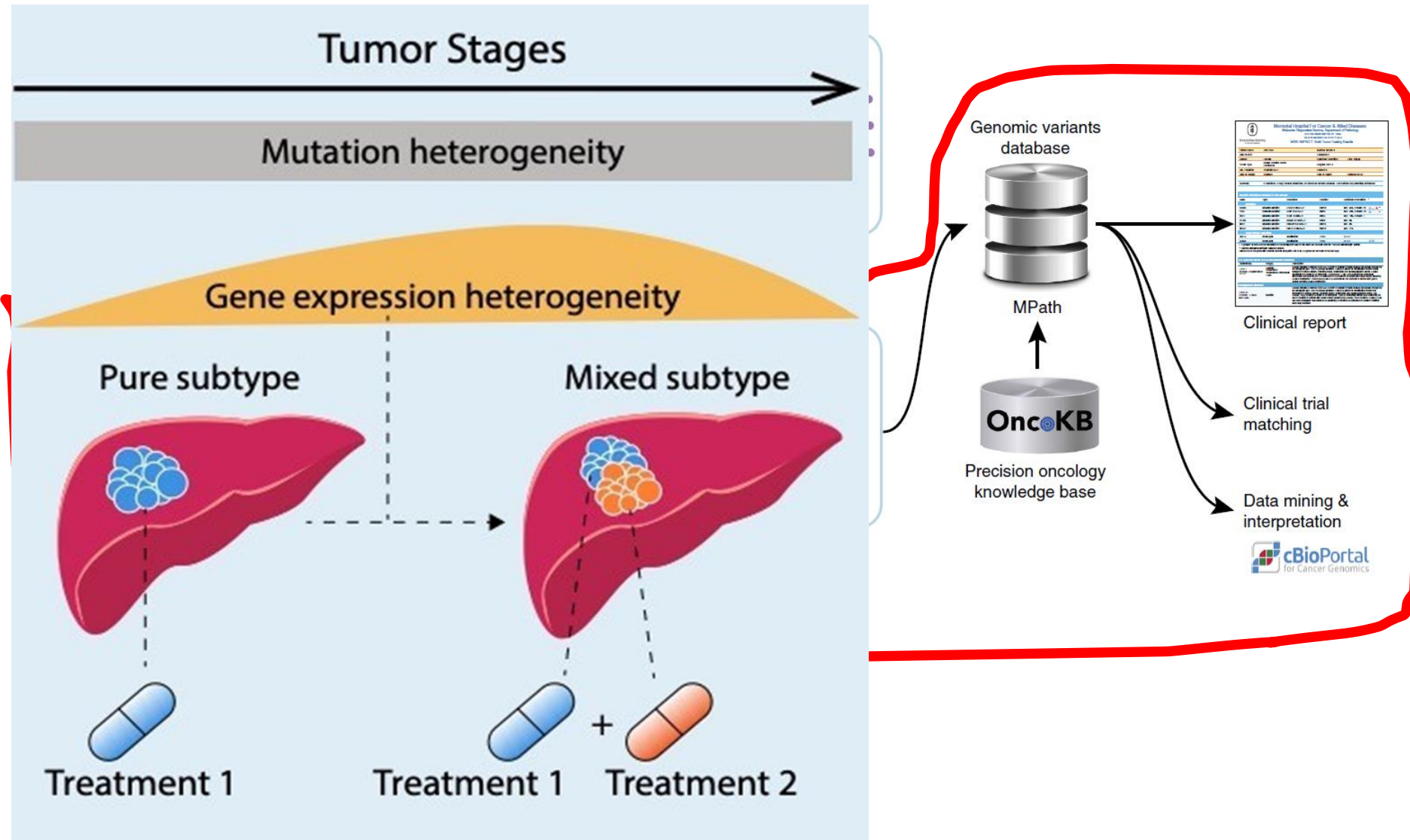
Esophageal cancer is an aggressive tumor that has a high rate of incidence and mortality worldwide. It is the 10th most frequent type in Brazil, being squamous cell carcinoma (ESCC) the predominant subtype. There is currently an incessant search to identify the frequently altered genes associated with esophageal squamous cell carcinoma biology that could be druggable. This study aimed to analyze the somatic mutation profile of a large panel of cancer-related genes in Brazilian ESCC. In a series of 46 ESCC diagnoses at Barretos Cancer Hospital, DNA isolated from paired fresh-frozen and blood tissue, a panel of 150 cancer-related genes was analyzed by next-generation sequencing. The genes with the highest frequency of mutations were *TP53* (39/46, 84.8%), followed by *NOTCH1* (7/46, 15.2%), *NFE2L2* (5/46, 10.8%), *RB1* (3/46, 6.5%), *PTEN* (3/46, 6.5%), *CDKN2A* (3/46, 6.5%), *PTCH1* (2/46, 4.3%) and *PIK3CA* (2/46, 4.3%). There was no significant association between molecular and patients' clinicopathological features. Applying an evolutionary action score of p53 (EAp53), we observed that 14 (35.9%) *TP53* mutations were classified as high-risk, yet no association with overall survival was observed. Concluding, this the largest mutation profile of Brazilian ESCC patients, which helps in the elucidation of the major cancer-related genes in this population.

Mutations analysis in esophageal Cancers

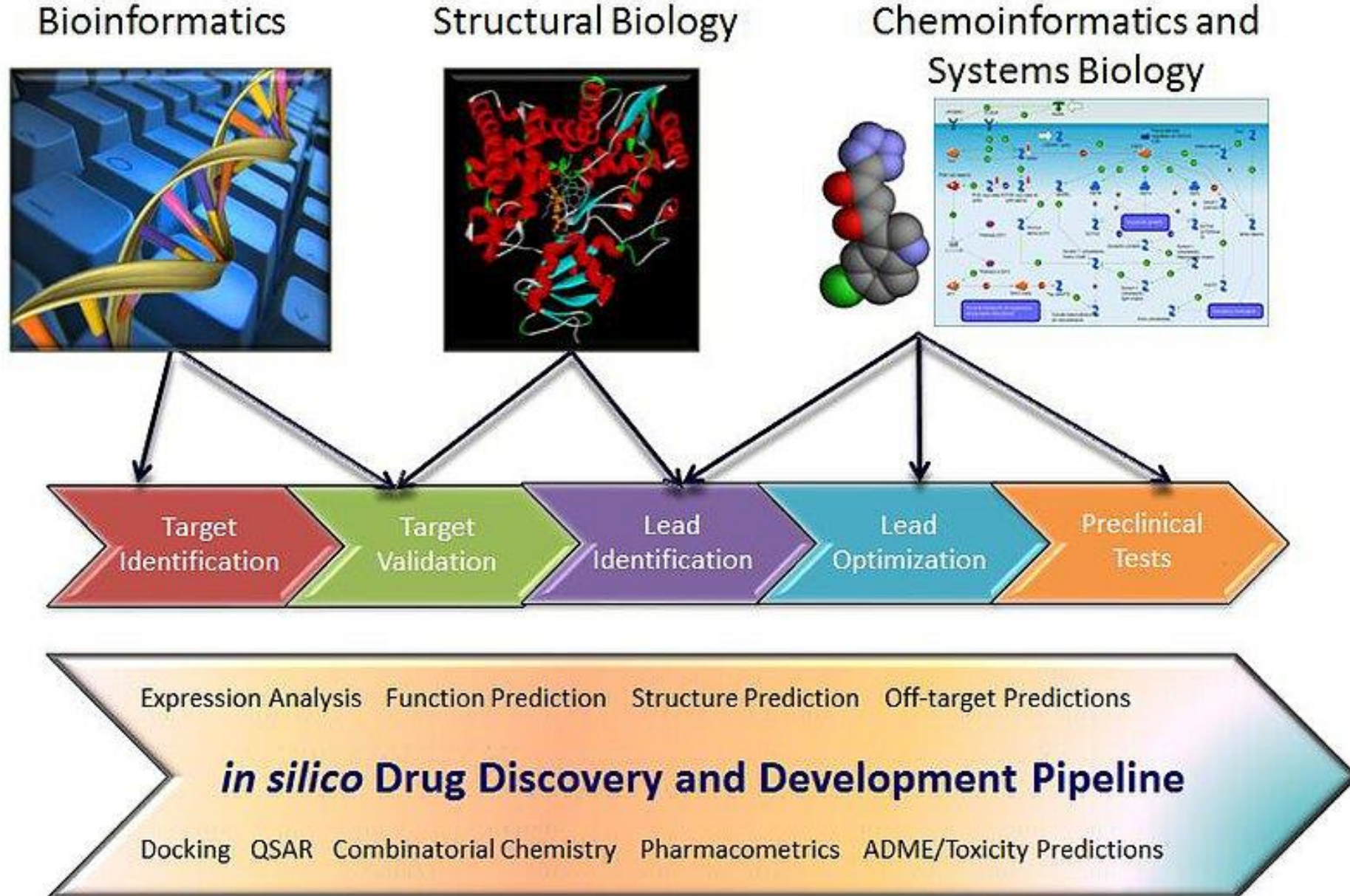


Do we really need to know this?

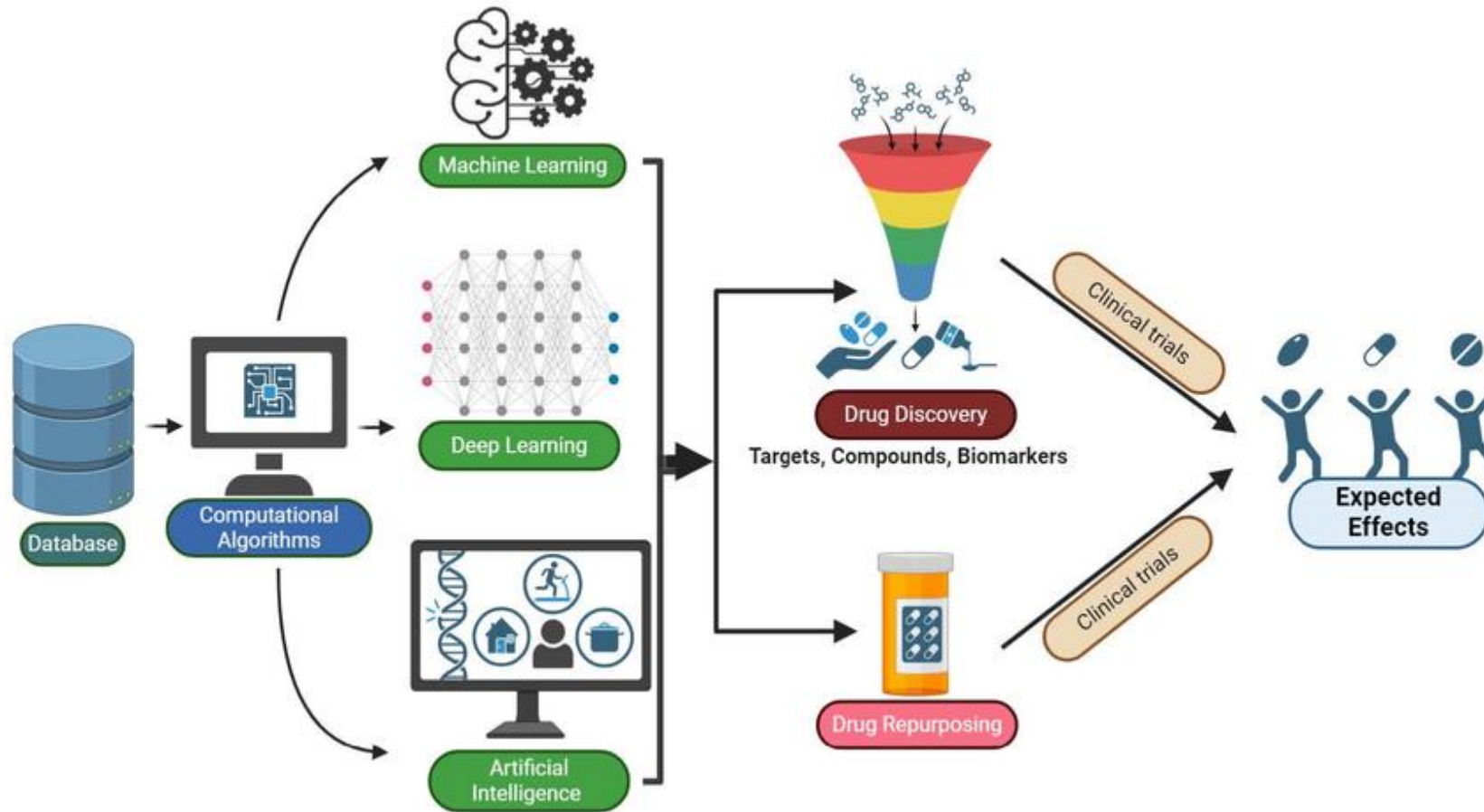
Schematic diagram for the Neoantigen target (mutations) discovery from Hepatocellular carcinoma (HCC) patients:



Computational Drug Discovery: Technologies and Methods

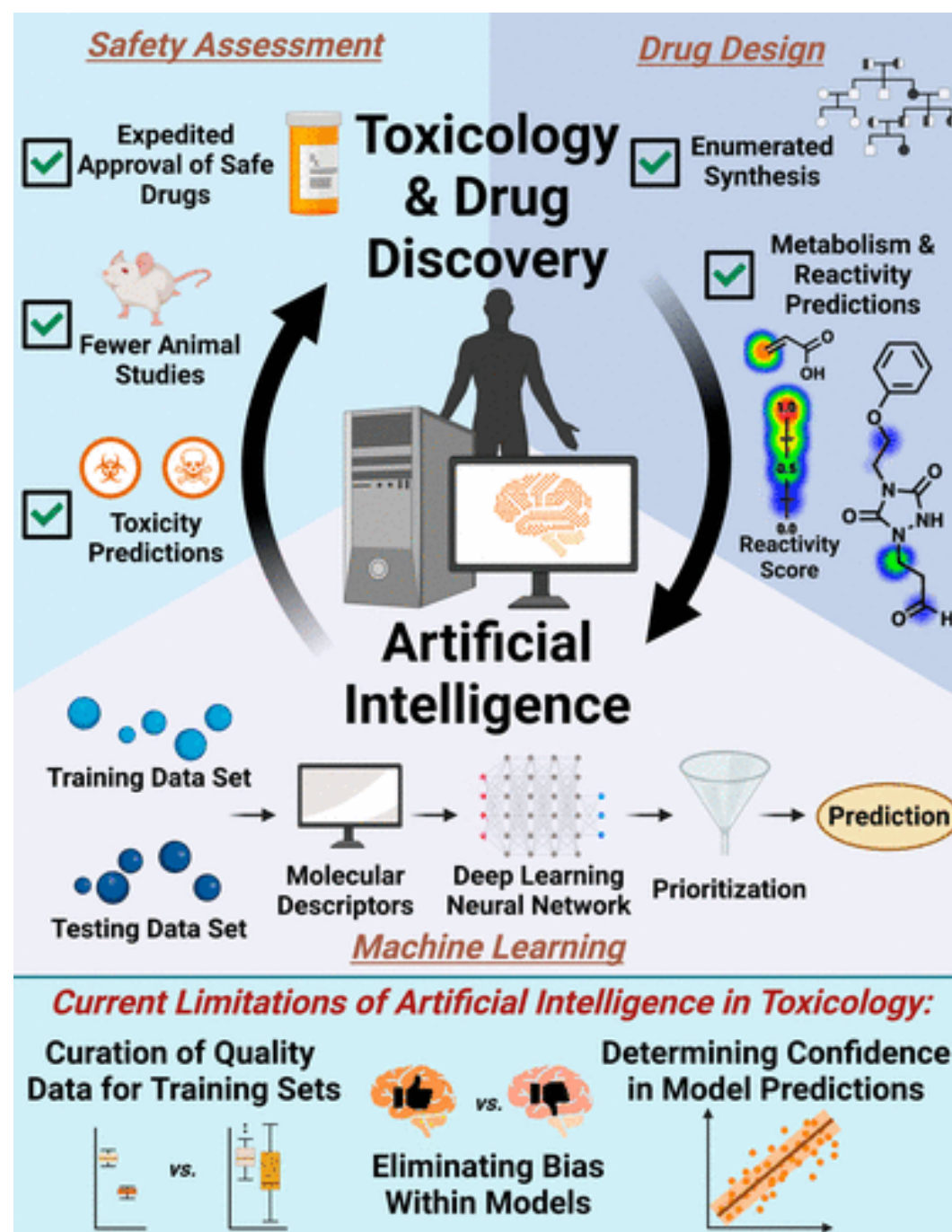


Machine learning (ML), Artificial intelligence (AI), and deep learning (DL) in drug discovery



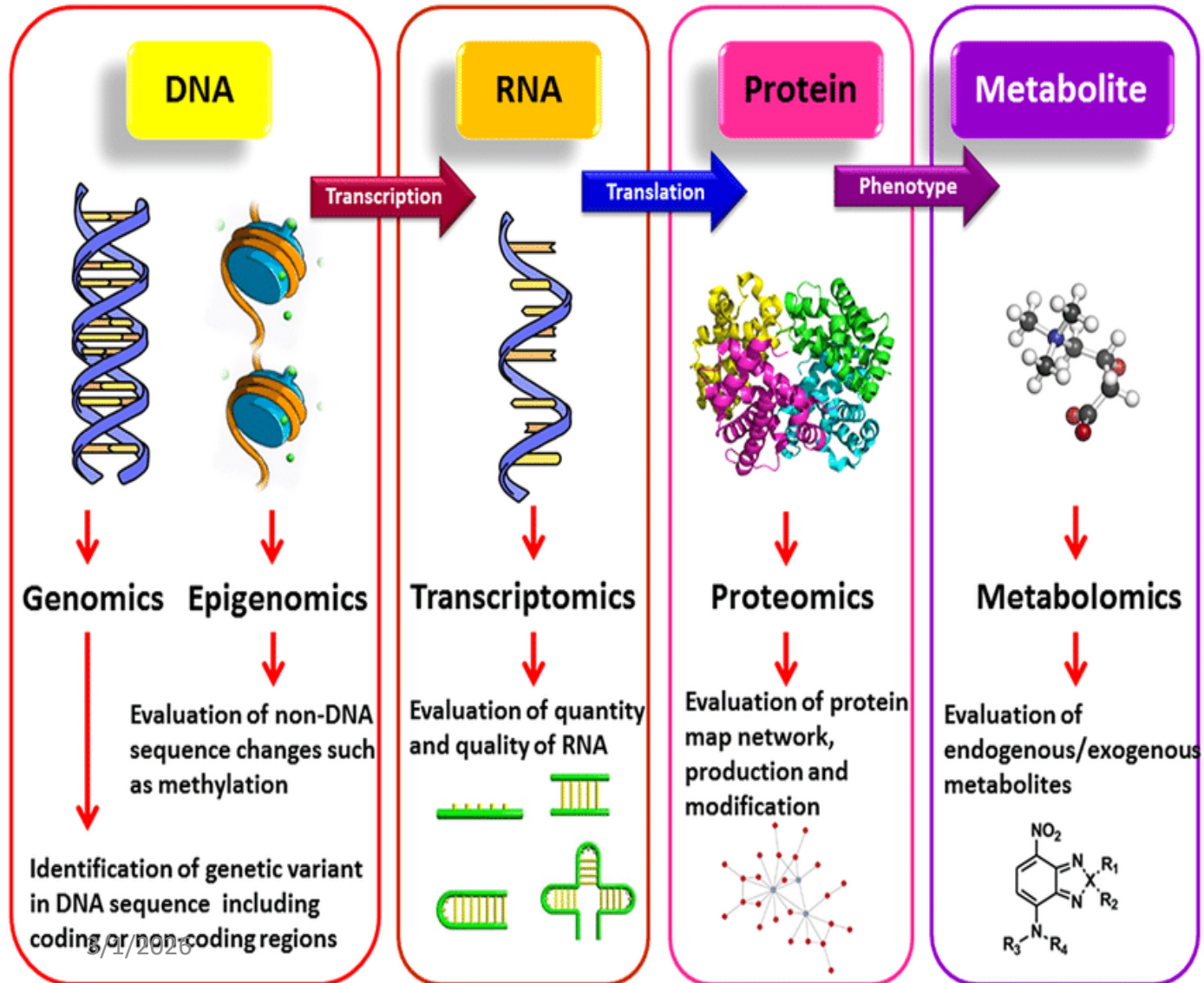
This schematic represents and development, including drug repurposing for cancer treatment. The diagram illustrates how computational algorithms analyze large datasets, leading to insights that inform different aspects of the drug discovery process, such as identifying therapeutic targets, compound screening, drug design, biomarker discovery, patient stratification, and clinical trial optimization.

AI is being applied in various steps of drug design, development, and safety assessment.



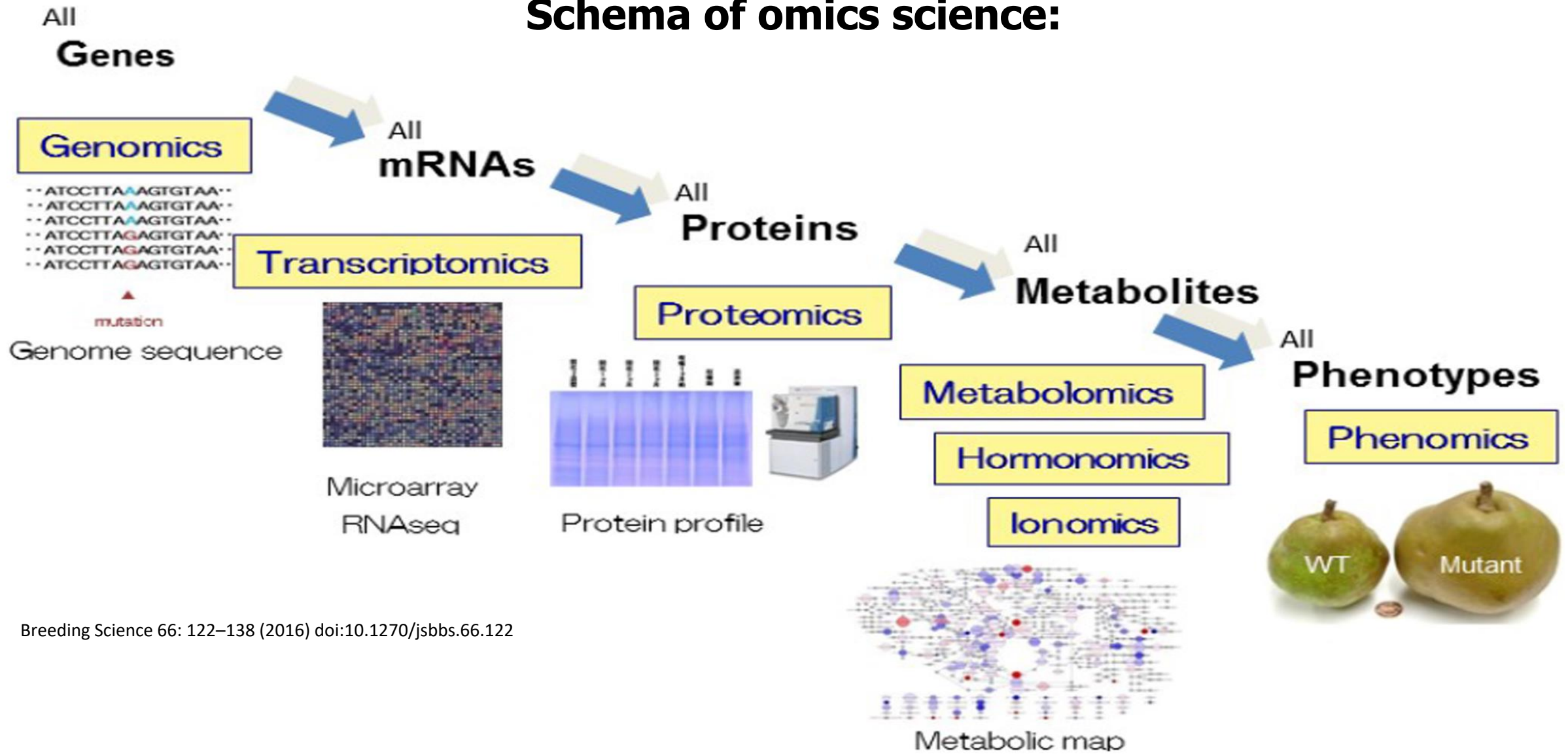
OMICS Science:

The omics sciences share the overarching aim of **identifying**, **describing**, and **quantifying** the biomolecules and **molecular processes** that contribute to the **form and function of cells** and tissues.



Different main levels of “-omics” technology for evaluation of comprehensive molecules in cell including genetic variants in DNA sequence (Genomics), non-DNA sequence alternation such as histone modification and methylation (Epigenomics), analysis of expression and structural changes in RNA and variants like splice sites (Transcriptomics), evaluation of expression, modification and net protein interactions (Proteomics) and description of functional metabolites in cell (Metabolomics). The mix of different type of “-omics” technology can help us in diagnose, prognoses and therapeutic approaches of tumors

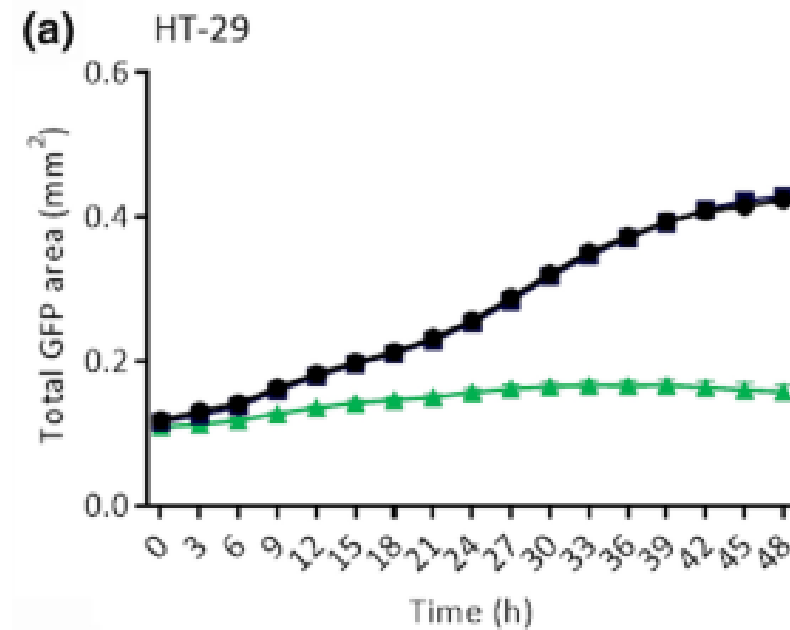
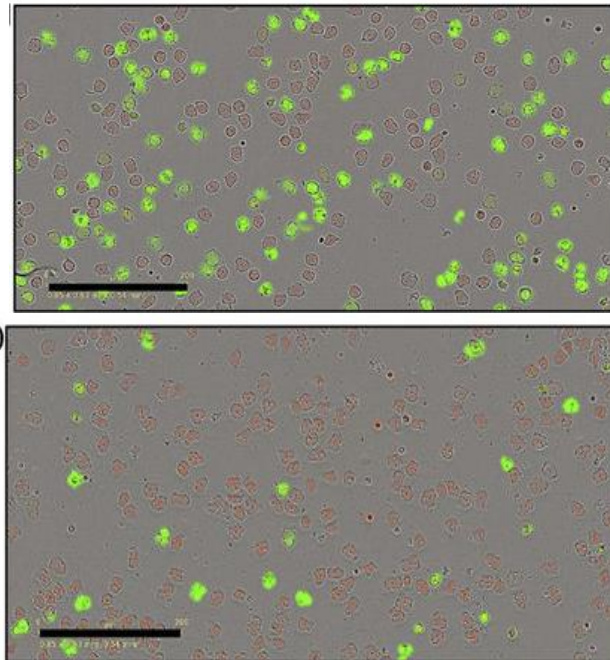
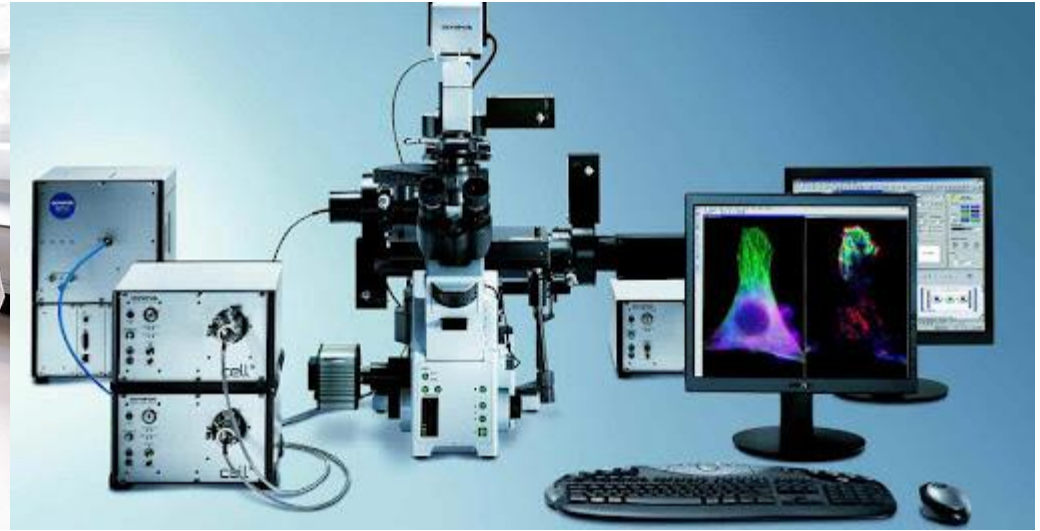
Schema of omics science:



Breeding Science 66: 122–138 (2016) doi:10.1270/jsbbs.66.122

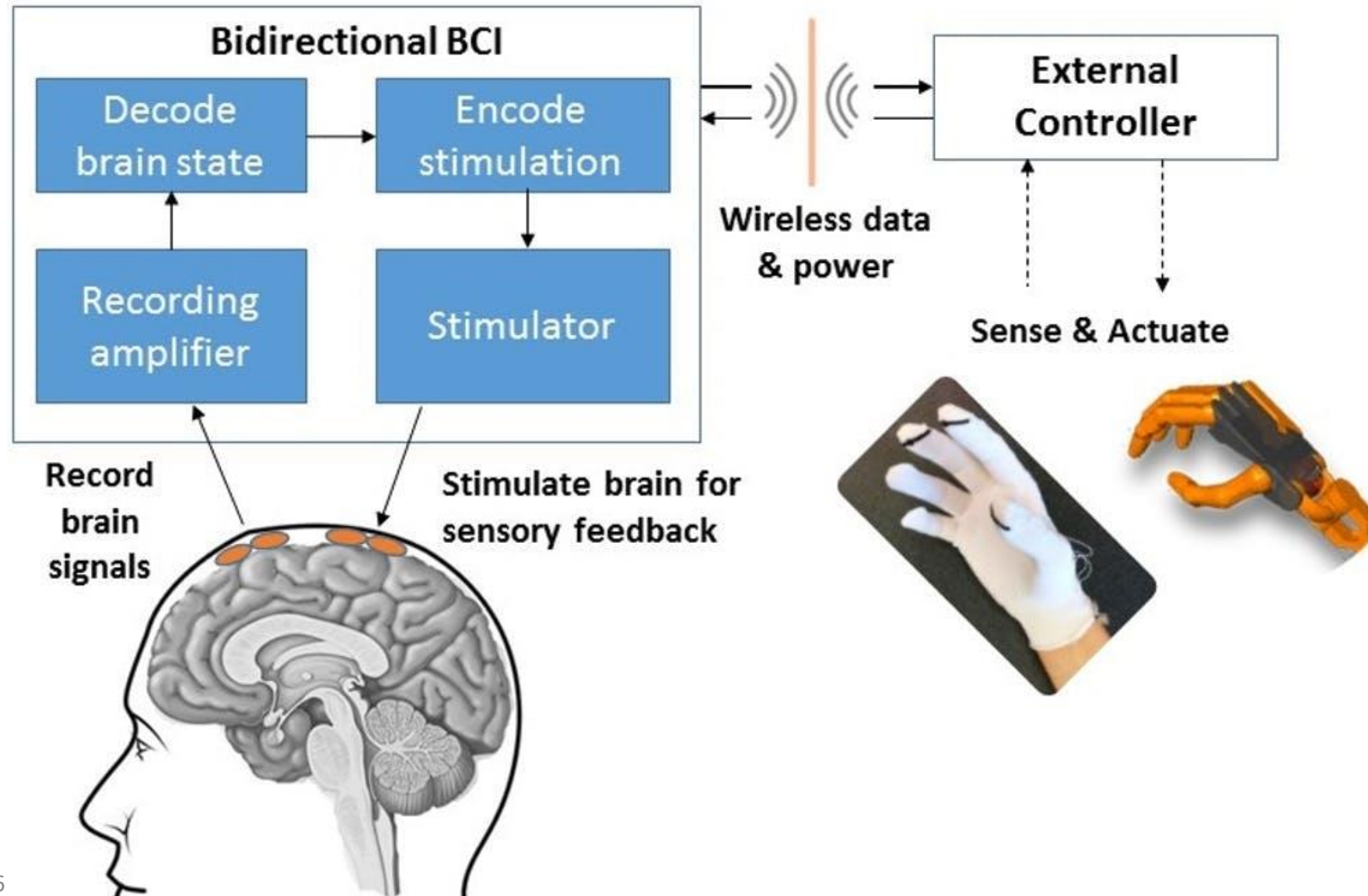
Comprehensive studies targeting all genes, transcripts (RNAs), proteins, metabolites, hormones, ions and phenotypes are called genomics, transcriptomics, proteomics, metabolomics, hormonomics, ionomics and phenomics, respectively.

Live Cell imaging:



A brain-computer interface (BCI)

Brain and Neuroscience

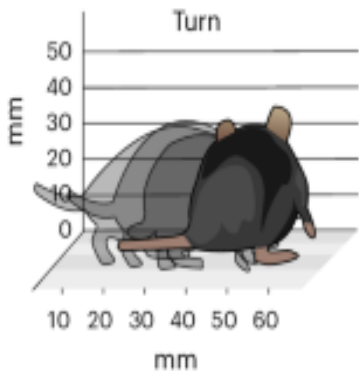
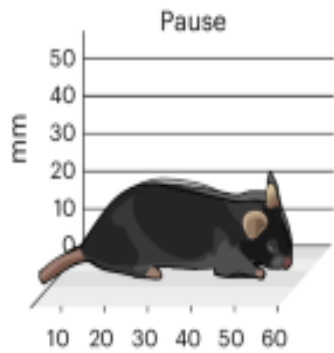
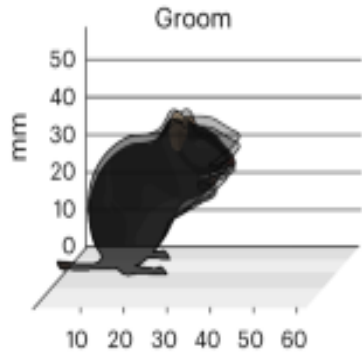


Behavioral Neuroscience

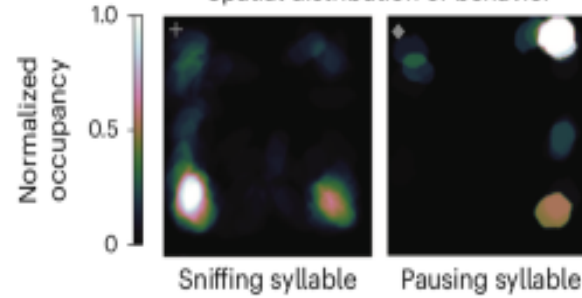
Behavior recording
with depth sensor



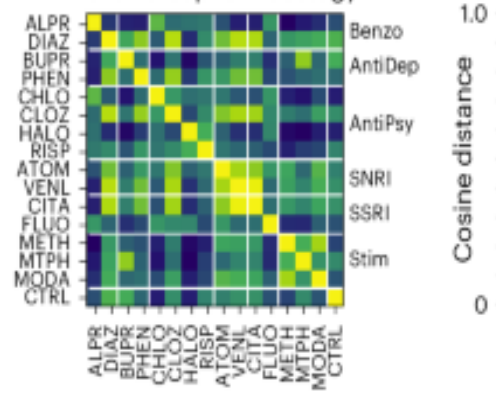
Discover syllables
from behavior



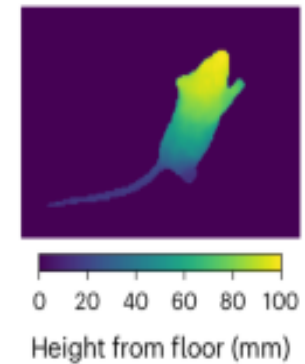
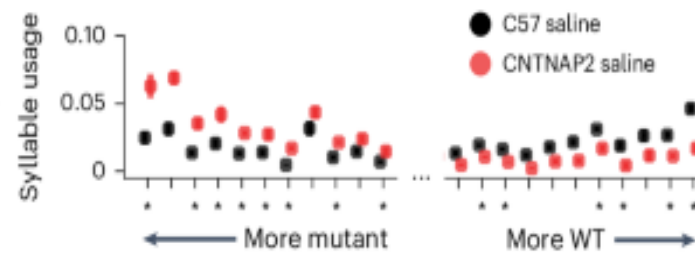
Spatial distribution of behavior



Behavioral pharmacology



Genotype-phenotype mapping

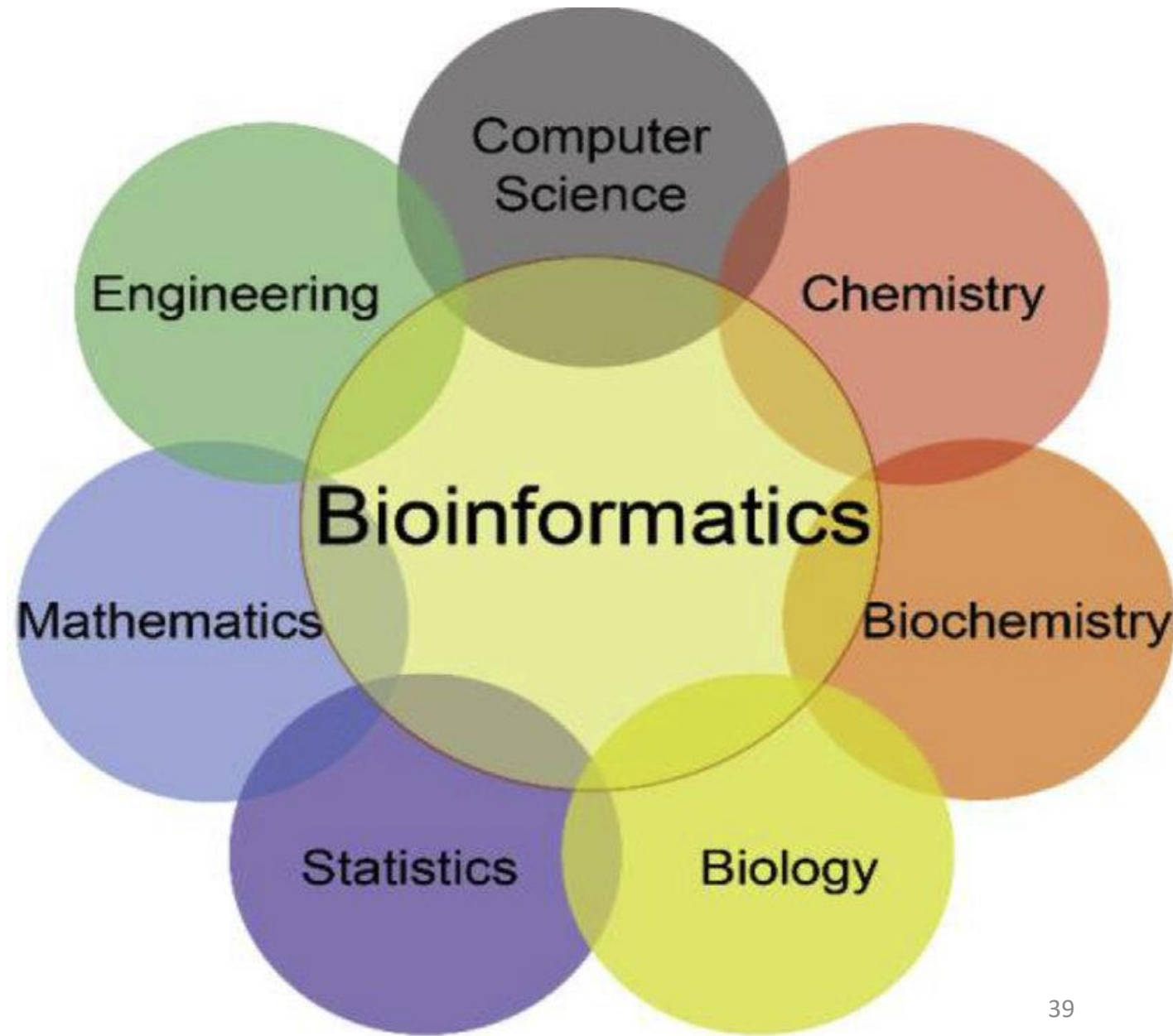


Height from floor (mm)

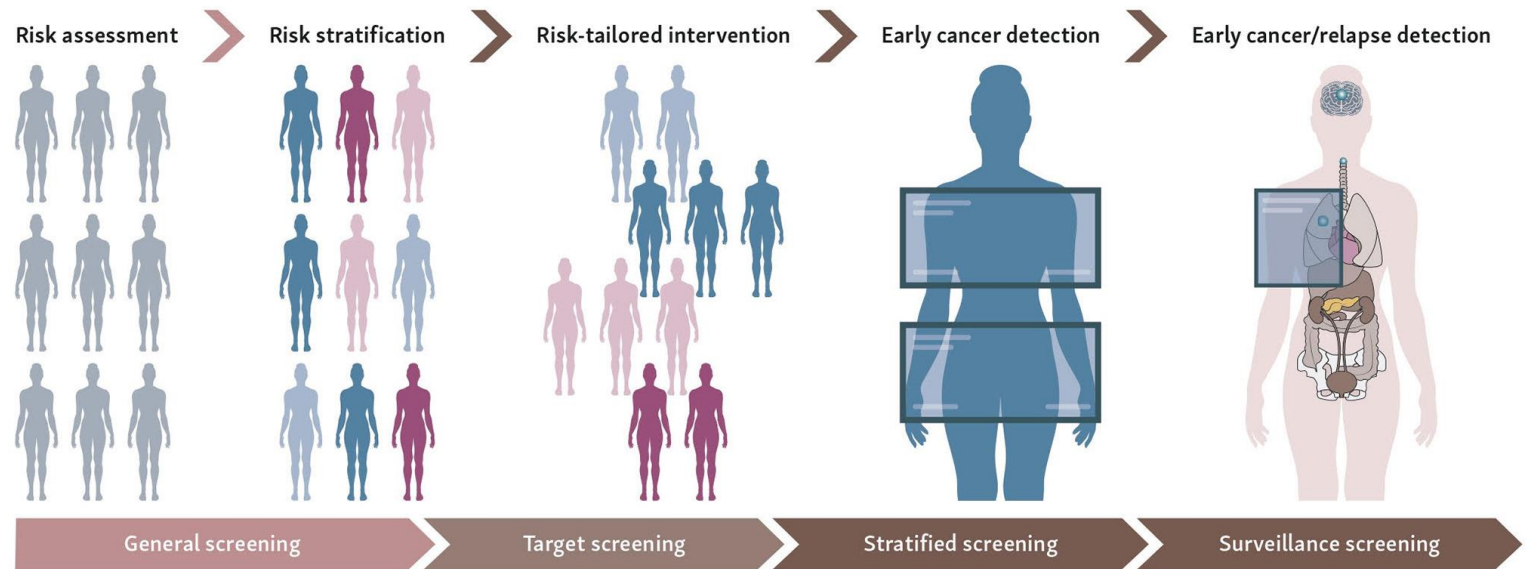
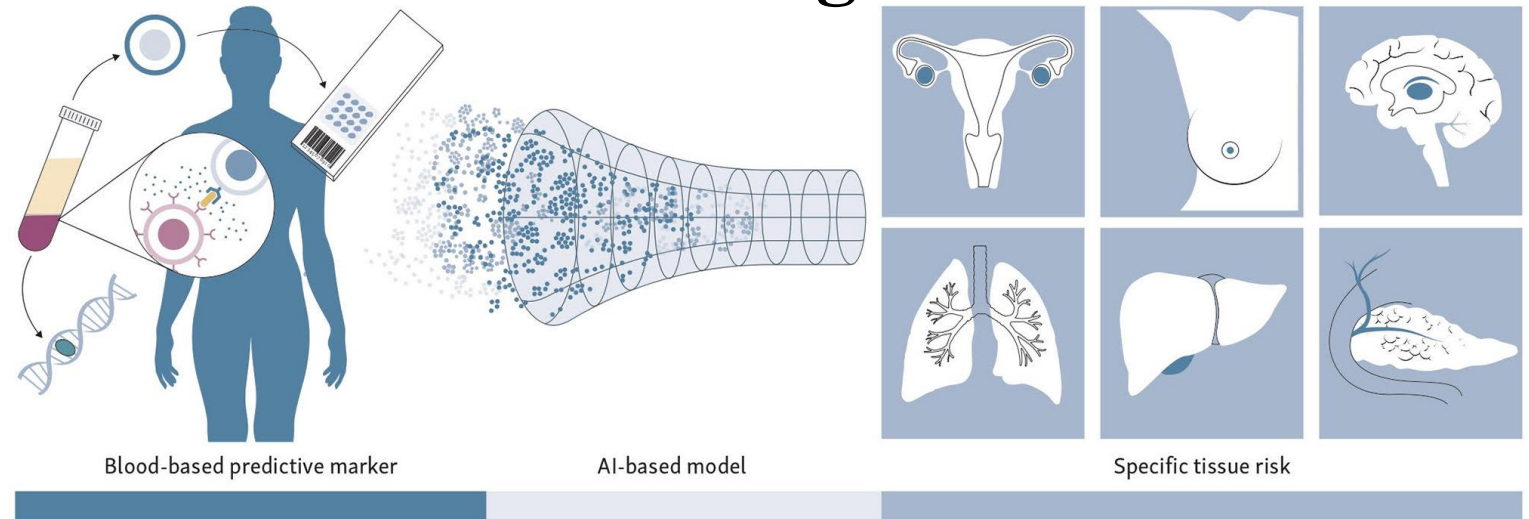
Introduction: Scope for Engineers

Bioinformatics, Systems Biology, and Computational Biology

These are currently popular fields of study which are highly multi-disciplinary, and engineers can significantly contribute to those fields. Those fields of study computationally analyse very large data sets to draw insights into the working of the fundamental functional unit of life—the cell. To make sense of such large data sets, people with backgrounds as diverse as computer science, biology, biochemistry, biological engineering, electronics engineering, and many others, need to come together.



Artificial intelligence for cancer screening and surveillance



ML algorithms

- to translate cancer genomics into precision medicine
- to digitalize cytological slides to fast-paced diagnostic processes

AI-system setting

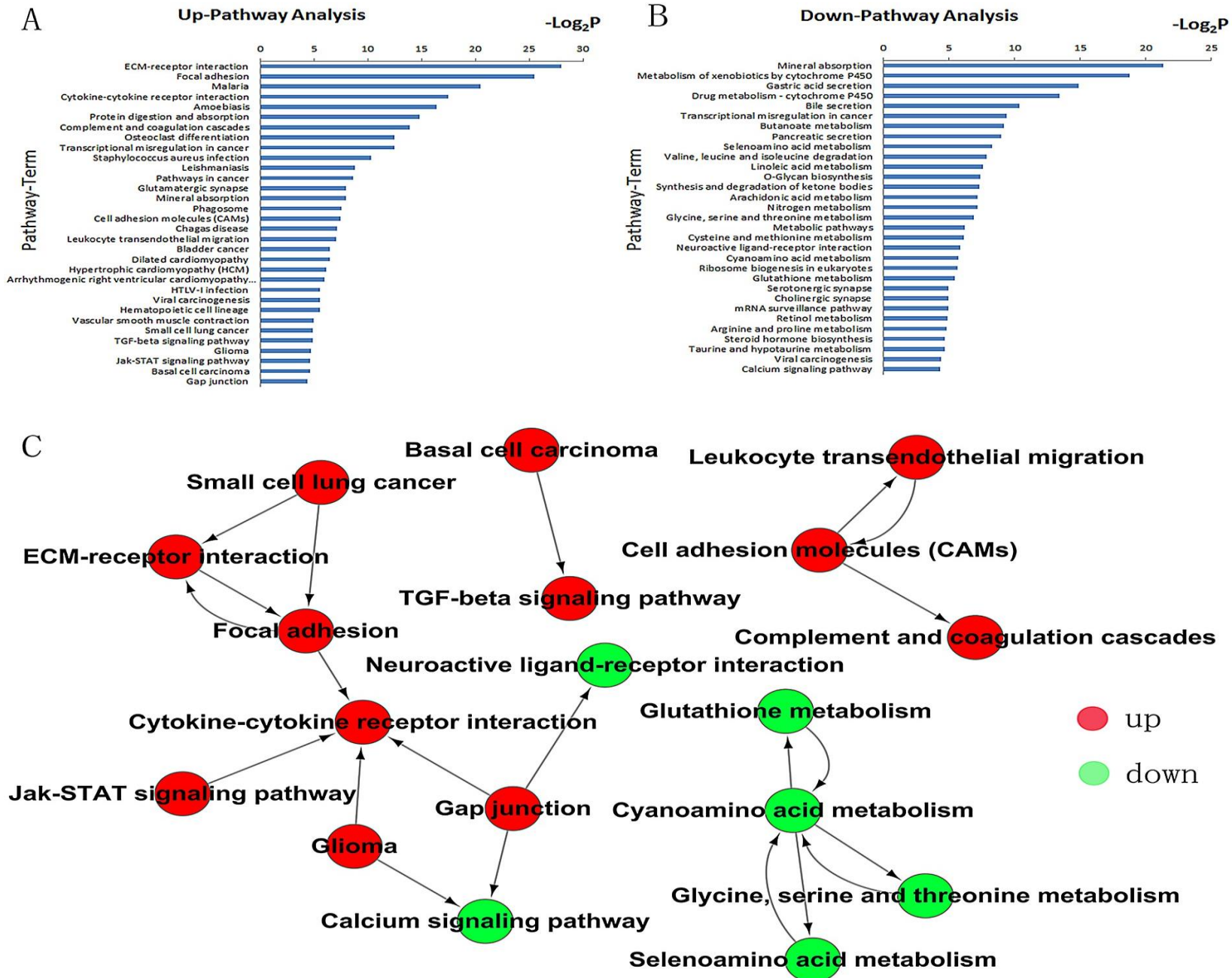
ML algorithms

- to outperform human in reading digital radiological images
- to assess blood-based predictive marker and radiology imaging

■ High risk ■ Moderate risk ■ Intermediate risk ■ Low risk

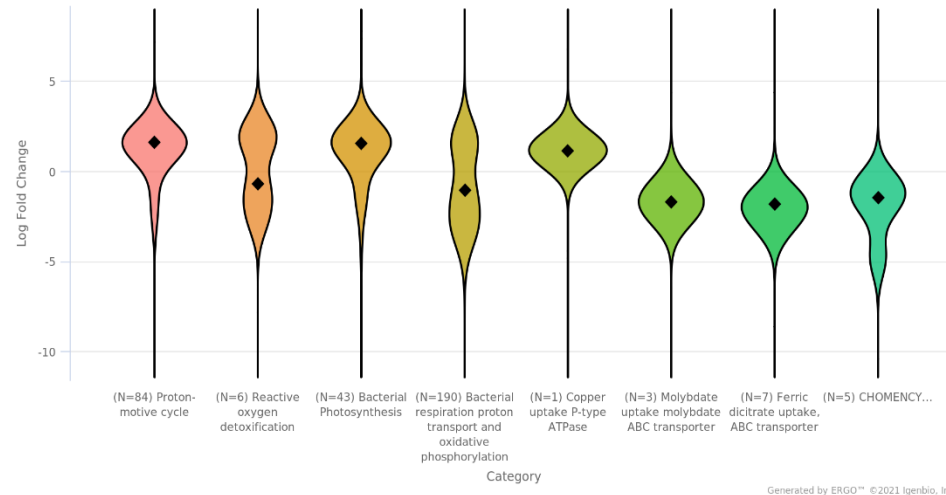
<https://www.esmorwd.org/article/S2949-8201%2824%2900024-9/fulltext>

Pathway analysis of differentially expressed genes

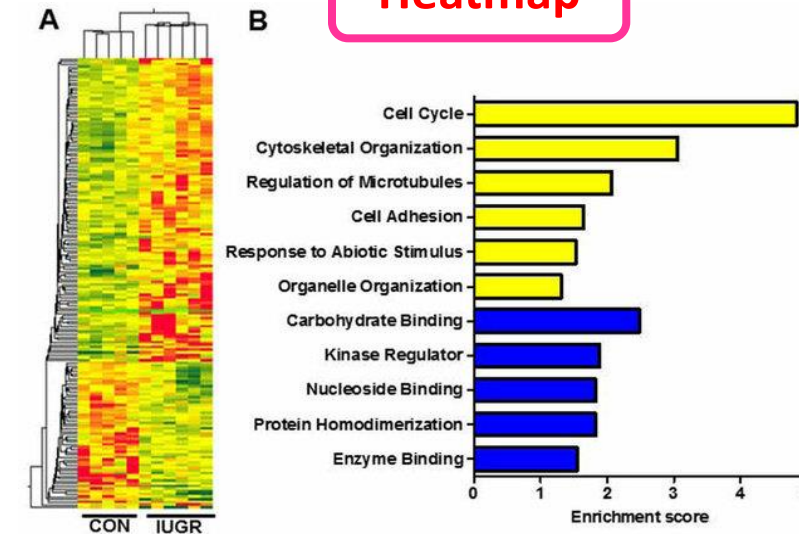


RNaseq analysis of differentially expressed genes

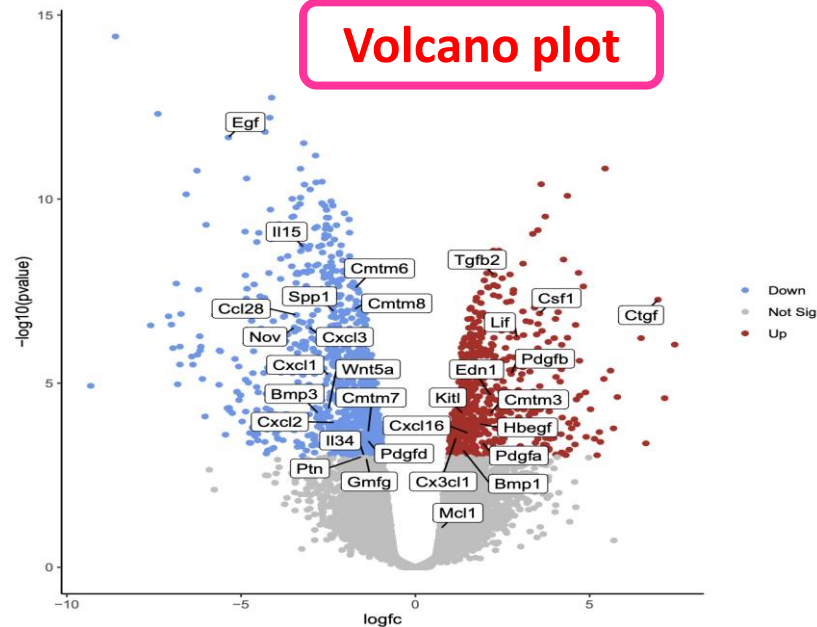
Violin Plot



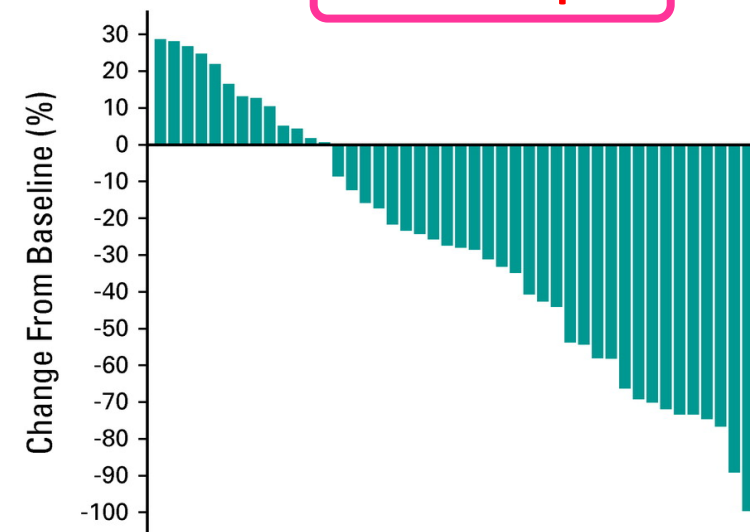
Heatmap



Volcano plot



Waterfall plot



Commonly used programming language in Biology:

Python

Versatile for RNA-seq data processing, analysis, and visualization using libraries like Biopython, Pandas, and matplotlib.

Biopython

A Python library for handling DNA/RNA/protein sequences, annotations, and biological file formats (FASTA, GenBank, etc.).

R

Excellent for statistical analysis and visualization of RNA-seq data, especially with Bioconductor packages like DESeq2 and edgeR.

MATLAB

Useful for modeling biological systems, signal processing in omics data, and image analysis in cell biology.

SAS:

Strong for statistical analysis of clinical and experimental biology data, including large-scale genomics studies.

Perl

Traditionally used for parsing and automating genome annotation and sequence data processing tasks.

Java

Powers bioinformatics tools like IGV and GATK, useful for variant analysis and genome visualization.

Julia

Emerging in computational biology for high-performance modeling and numerical analysis of large-scale omics data.

Bioconductor

An R-based ecosystem for comprehensive RNA-seq analysis—normalization, differential expression, visualization, and more.

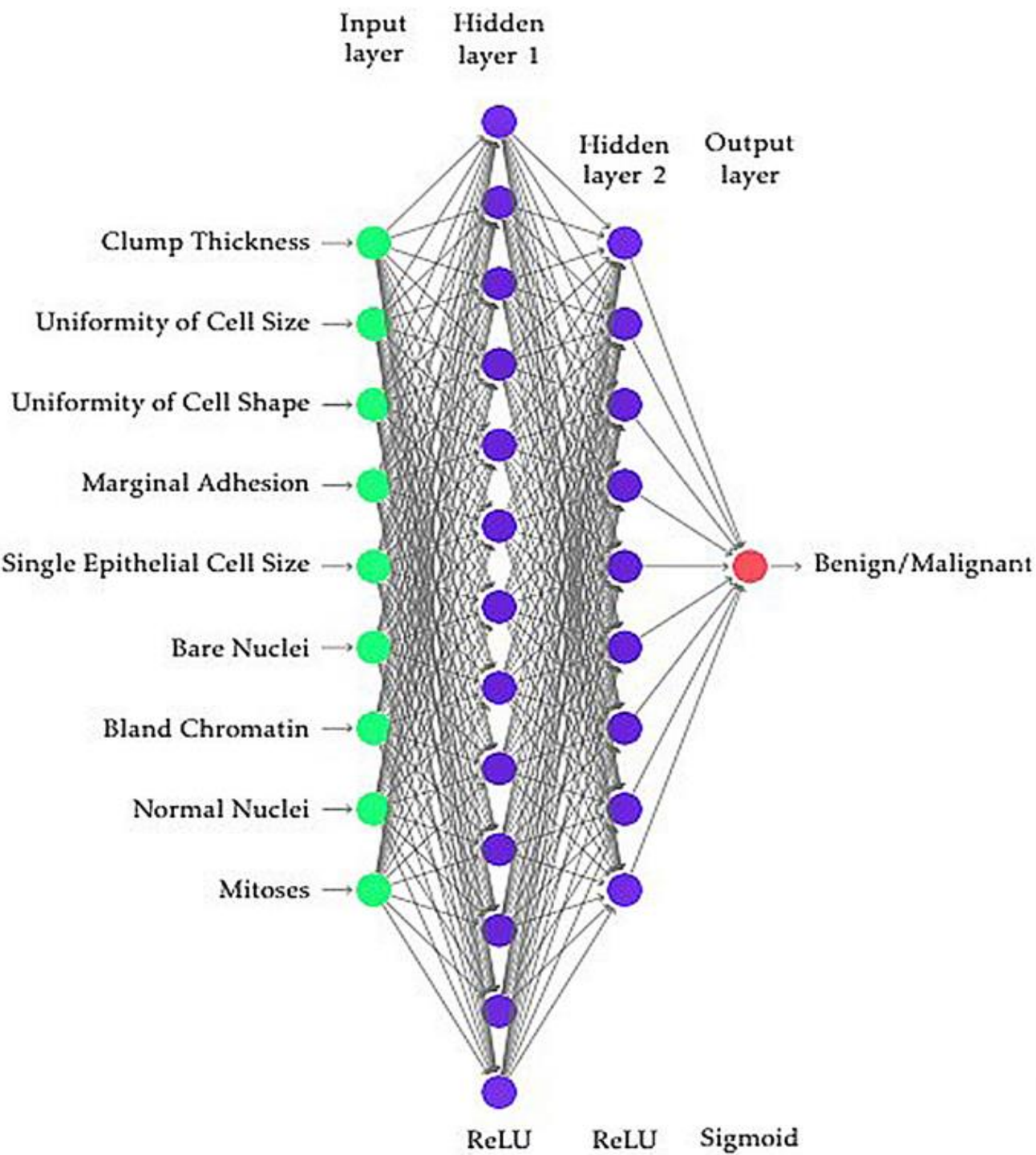
SQL

Efficient for managing and querying large biological databases, such as gene expression or sequencing metadata.

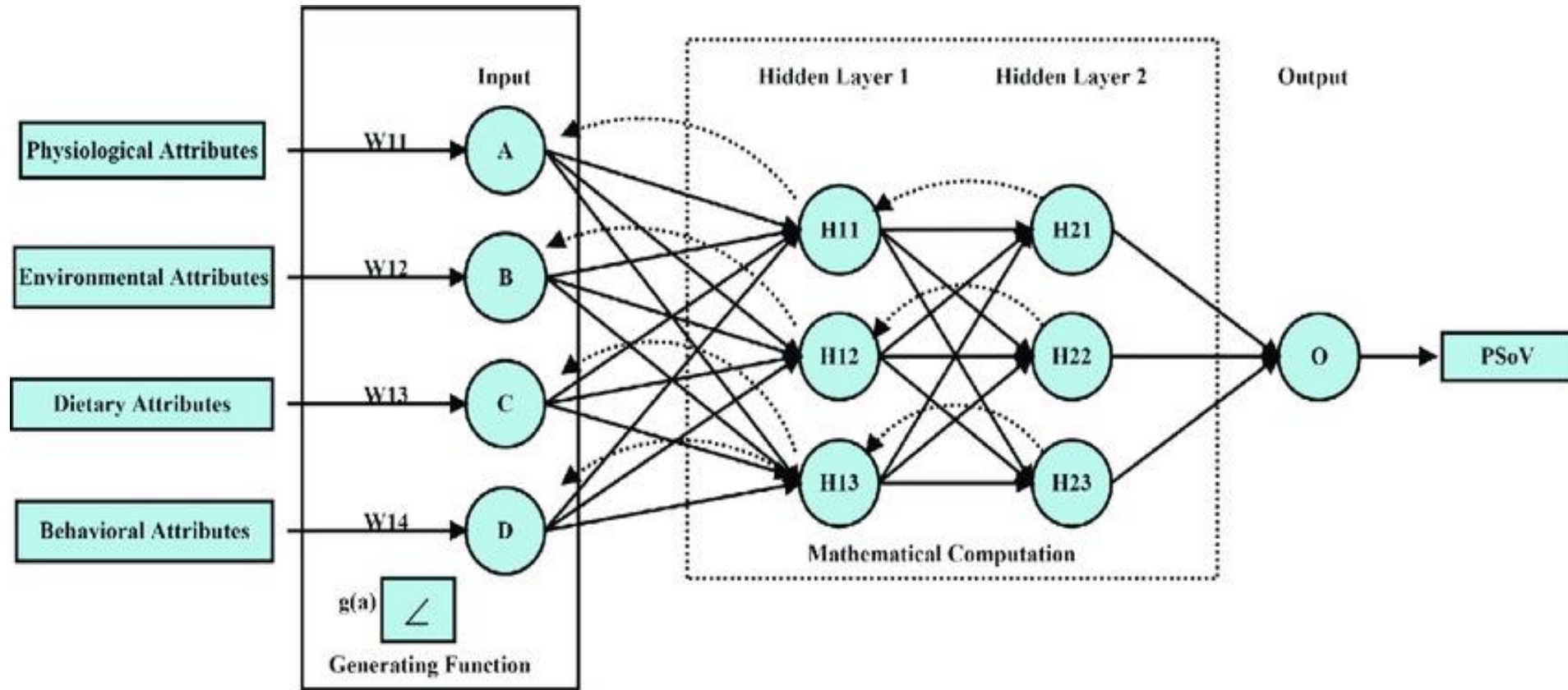
Breast Cancer Prediction: A Comparative Study Using Machine Learning Techniques

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Artificial Neural Network



Artificial Neural Network The artificial neural network (ANN), a processing set-up, is supposedly inspired by the working of animal brains—they learn by example. ANNs are made up of a large number of interconnected elements, each of which work similar to a biological nerve cell. ANNs can be used to derive useful information from imprecise data. For example, they can be used to detect patterns or trends in complicated data that are difficult to detect by manual inspection.

Popular neural network architectures and models used in biology, bioinformatics

Artificial Neural Network (ANN)	Basic neural network model, useful for pattern recognition and prediction.
Deep Neural Network (DNN)	Multi-layer ANNs for modeling complex relationships in large biological datasets.
Convolutional Neural Network (CNN)	Great for analyzing biological images (e.g., microscopy, histology) and genomic sequences.
Recurrent Neural Network (RNN)	Ideal for sequence data such as DNA, RNA, and protein sequences.
Long Short-Term Memory (LSTM)	An advanced RNN used for modeling long-range dependencies in biological sequences.
Transformer	State-of-the-art architecture for sequence modeling (used in AlphaFold, protein folding, etc.).
Autoencoder	Used for dimensionality reduction and noise filtering in omics data.
Variational Autoencoder (VAE)	Useful in generating synthetic biological data and modeling latent biological features.
Generative Adversarial Network (GAN)	Used for generating synthetic biological images or sequences.
Graph Neural Network (GNN)	Effective for modeling biological networks like protein-protein interactions or gene regulation networks.

JobSites

<https://biotech-careers.org/articles/top-3-best-computer-science-jobs-biotechnology>

<https://www.indeed.com/jobs?q=bioinformatician&l=USA&from=searchOnHP&vjk=868cf1b9f153f161>

<https://www.indeed.com/jobs?q=Computer+Science+%26+Biology&l=&ts=1730695239729&from=searchOnHP&rq=1&rsIdx=1&fromage=last&vjk=eb5b1becb936467d>

<https://www.indeed.com/jobs?q=Biology+and+Computer+Engineer&l=&from=searchOnHP&vjk=1c5b9f335bce570c>